

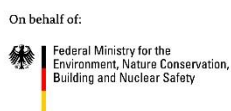


**ECAS INSTITUTE IN COLLABORATION WITH THE INSTITUTE FOR CLIMATE
CHANGE AND ADAPTATION (ICCA)**

**Baseline Survey for ‘Operationalizing Green Economy Transition in Africa: Status
of Green Economy initiatives in Kenya’**

Mombasa County Findings Report

January 2016



of the Federal Republic of Germany



This project is part of the International Climate Initiative (IKI). The German Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BMUB) supports this initiative on the basis of a decision adopted by the German Bundestag.

1. Overview

Mombasa County is located in the central-south portion of Kenya's Indian Ocean coastline. It covers an area of 229.9 Km² excluding 65 Km² of water mass which is 200 nautical miles into the Indian Ocean, and borders Kilifi County to the North, Kwale County to the South West and the Indian Ocean to the East (Mombasa County Annual Development Plan, 2014). The County lies between latitudes 3°56' and 4°10' South and between longitudes 39°34' and 39°46' East (Mombasa County Annual Development Plan, 2014). It has a population of about 1 million, an annual population growth rate of 3.45 % (1999-2009), and a population density of 4,289 people per km² based on the 2009 national population census (Commission on Revenue Allocation, 2013).

1.1. Biophysical Setting

The County lies within the Coastal lowland which rises gradually from the sea level in the East to about 132m above sea level in the mainland, and is characterised by three distinct physiographic features, which include the coastal plain, the hilly areas mainly found within the Western part of the County, and the shoreline-ocean zone with fringing coral reefs and tidal flats (Mombasa County Annual Development Plan, 2014).

A synopsis of the climate is given in the Mombasa County Annual Development Plan, 2014. The report notes that the rainfall pattern is characterized by two distinct long (April - June) and short (October to December) seasons, with an annual average of 640mm. The annual mean temperature in the County is 27.9°C with a minimum of 22.7°C and a maximum of 33.1°C. The hottest month is February with a maximum average of 33.1°C while coolest is July with a minimum average of 22.7°C, and the average humidity at noon is about 65 per cent.

The County's ecosystem has both marine and terrestrial components: both ecosystems are characterized by diverse species of flora and fauna, the most common being coconut trees and different species of fish, which have different cultural, social and financial values (Mombasa County Annual Development Plan, 2014). The fringing coral reefs, creeks and tidal flats with extensive mangrove forests are breeding grounds for fish, while the fringing coral reefs in North Coast are an important marine conservation area hosting the Mombasa Marine National Park and Reserve (Mombasa County Annual Development Plan, 2014).

1.2. Socio-economic Setting

The coastal region is a key contributor to the National Gross Domestic Product with 45% from tourism and 15% from the shipping activities which are important to the coastal economy (WRI, 2007; Government of Kenya, 2009). The main economic activities include tourism,

shipping, commerce and subsistence farming (Mombasa County Annual Development Plan, 2014). The community's sources of livelihood include: Small scale fisheries, marine fisheries, mariculture in community based farms, tourism, agriculture and forestry, mineral processing such as oil and gas, cement manufacturing at Bamburi Cement, coral limestone, shale and sand mining (Mombasa County Annual Development Plan, 2014). At 60 per cent, the formal sector provides majority of employment, and the major employers are the hotel industry, shipping industry, Government of Kenya and various private institutions (Mombasa County Annual Development Plan, 2014). It was noted some years ago that Mombasa should be a prosperous city, offering plentiful economic opportunities to its population, but that it had not yet realised its potential since services and infrastructure were becoming increasingly inadequate and about a quarter of its households fell below an absolute poverty line (Rakodi et al., 2000).

The population distribution and settlement patterns in the County are influenced by proximity to vital social and physical infrastructure networks such as roads, housing, water and electricity (Mombasa County Annual Development Plan, 2014). Other factors that influence settlement patterns include accessibility to employment opportunities, and security. Despite efforts being made to settle people, the County still has a very large number of landless people most of whom live in the city's slums of Mishomoroni, Junda and Kisumu ndogo in Kisauni Sub-County, Shika-Adabu and Ngomeni in Likoni Sub-County, and Bangladesh in Changamwe Sub-County (Mombasa County Annual Development Plan, 2014). The land adjudication process is ongoing for Shika-Adabu and Vyemani settlement schemes, but there are also other proposed schemes in the County namely; Maweche, Kibundani, Ujamaa-Shonda and Kidungunyi (Mombasa County Annual Development Plan, 2014). There sparsely populated areas in the outskirts of the County, which include Mwakirunge-Maunguja, Mwangala, Mreroni and the Mkupe Jetty area, are least developed in terms of infrastructure such as road network, electricity and water supply (Mombasa County Annual Development Plan, 2014). Education and health facilities are also scantily available in these areas making the inhabitants highly prone to poverty and disease incidences (Mombasa County Annual Development Plan, 2014).

In Mombasa County, 28% of the residents with no formal education, 37% of those with primary level of education and 46% of those with secondary level of education or above are working for pay (Ngugi, 2013) (Table XX). Mombasa County's Gini index is 0.365 compared with Turkana County, which has the least inequality nationally (0.283) (Ngugi, 2013).

Table XX. Overall Employment by Education Levels in Mombasa County (Ngugi, 2013)

Education Level	Work for pay	Family Business	Family Agricultural Holding	Intern/ Volunteer	Retired/ Home-maker	Fulltime Student	Incapacitated	No work	Number of Individuals
Total	41.0	15.7	3.1	1.3	15.5	10.0	0.3	13.1	598,021
None	28.4	15.5	6.8	2.6	28.2	1.1	1.2	16.3	56,308
Primary	37.3	16.3	3.4	1.1	19.1	8.0	0.3	14.6	239,944
Secondary+	46.2	15.2	2.2	1.3	10.3	13.3	0.1	11.4	301,769

2. Sectoral Analysis

During the inception stakeholder workshop held in Mombasa on the 14th October 2015, the stakeholders prioritized the following sectors for Green Economy initiatives: Waste management, Tourism, Cities and built environment, Transport and infrastructure, and Fisheries. Following further consultations with County officials, as well as the views of respondents to the questionnaire survey, it was agreed that Energy sector should also be included in the analysis.

2.1. Waste Sector

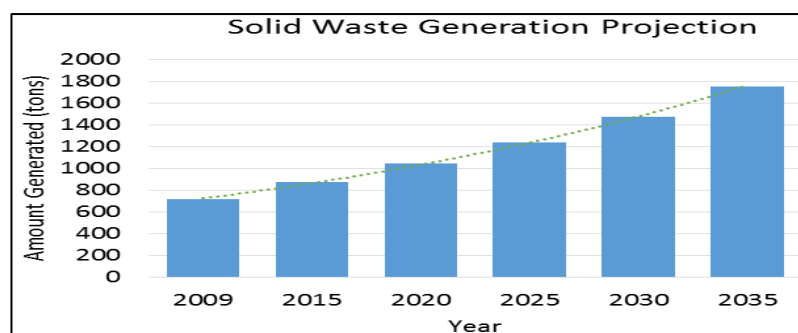
Mombasa County's solid waste generation rates and types are influenced by economic development, the degree of industrialization, public habits, local climate and the presence of industries like tourist (NEMA, 2016). Organic wastes form the largest component of the total amount of solid waste generated, contributing 68.35 percent; others are - textiles (2.06 %), paper (7.73 %), plastics (hard, soft and fiber) (12.6 %), metals (0.59 %), diapers (6.72 %) and glass (1.95 %) (NEMA, 2016). Mombasa County has been struggling to establish an efficient waste management system, partly due to difficulties in finding land for waste disposal and stakeholder resistance to establishment of a new dumpsite or landfill.

a) General economic sector indicators

There is generally no data available on the GDP of the waste sector nor on the contribution of the growth of the sector to the GDP at national or county level. The respondents, however, felt that the sector contribution to the County GDP and employment was, at the very least, important.

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Important to very important
Economic significance - Employment	Slightly important to quite important
Key sources of energy for the sector	Electricity, fossil fuels, solar power

The amount of solid waste generated annually is about 875 tonnes (Palfreman and Clark, 2015; Mombasa County Government, unpublished data), and the average waste per capita is estimated at 0.93 kg (Palfreman and Clark, 2015). It is projected that the amount of waste produced will rise to 1,735.87 tons per day by the year 2050, based on a projected population of about 2.3 million and using a conservative per capita generation rate of 0.76 kg per person per day (World Bank, 2012). NEMAs State of Environment Report for Mombasa County notes that presently, the ratio of organic to non-organic waste in low-income residential areas is 3:1, while in high income areas it is 2:1.



Source: draft ISUDP report, Mombasa County Government.

The households in Mombasa generate about 337 million litres of waste a day but the town faces a major challenge in disposal as only residents of Changamwe mainland have access to the service. Less than 300,000 of Mombasa's one million residents have access to sewerage services (CWSB, 2012). Mombasa has two sewerage treatment plants of Kizingo and Kipevu, but it's the latter that is fully functional after undergoing rehabilitation at KSh 297 million in 2014. Due to lack of a functional sewerage system on the island, some residents empty effluent into the Indian Ocean. The only functional sewage treatment plant at Kipevu serves residents of Changamwe, Port Reitz, Magongo and Jomvu.

b) Environmental issues and related indicators

Mombasa County has 3 dumping sites, namely, Kibarani, Mwakirunge and Likoni dump sites. Kibarani is located 7 km from the CBD and occupies approximately 8 acres of land in Mainland West. There are as yet no studies that have quantified CO₂ and methane emissions from both solid and liquid waste streams in the county. About 460 tons (53 percent) of the solid waste is handled by MOWASCO while the rest (47 percent) is informally disposed of through methods such as burning, burying and illegal dumping, thus contributing to environmental pollution (NEMA, 2016). It is estimated that 23 per cent of the waste is recycled or incinerated by commercial industries for export, while 16 per cent and 2 per cent are used for energy and heat recovery, respectively (Palfreman and Clark, 2015). Kibarani, the largest dumping site in the County, receives about 90 percent of all formally collected waste which averages about 386.5 tonnes. Mwakirunge and Likoni (Ujamaa Shonda) dumping sites receive 15.65 and 18.22 tonnes respectively (NEMA, 2016). About 55 percent of the solid waste that is formally managed comes from the island (draft ISUDP).

There are 13,286 of the 183,540 households who rely almost wholly on groundwater for their potable water needs, distributed as follows: wells 6,245 households, boreholes- 6,941 households (GOK, 2011). There is thus need for better management of solid and liquid wastes, including provision of adequate sanitation facilities and waste treatment plants, to ensure that such water is not contaminated.

Respondents (Planners)	Views	
Indicators		Remarks
Renewable energy used		None, but there was one mention for biogas production at Kongowea by a self-help group with the possibility of rolling this out to other area.
Resource efficiency - energy use		Not efficient to slightly efficient
Resource efficiency - products/materials recycled		Plastics, metals, bottles. Uncontrolled recycling of metals and plastics at the dumpsite (scavenging). Water (end product released to the ocean) Sludge from sewage used for manure
Waste water treatment and management methods		Conventional sludge activated system in Kipevu (Mainland) Rehabilitation of the Kizingo underground waste management plant on the Island Likoni (south), Nyali and Kisauni (north) use septic tanks and pit latrines Collected and transmitted to the treatment facilities Separation of properties such as: soil and silts (preliminary), aeration process (bacterial breakdown), sedimentation (sludge) and water.
Ecolabelling of products from the sector		None/Not aware of any being eco-labelled.

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Indiscriminate dumping</i>	-Blocked drainages leading to flooding, diseases, displacement	-Negative impacts on the budget -Non-compliance to service delivery	-Capacity building for environmental committees, MCAs, Senior Staff -Awareness creation on proper disposal methods	-No laws for enforcement -Enforcement -No designated landfill
<i>Ocean pollution</i>	-Decrease in fish stocks due to damage of breeding grounds -Loss of work and income from fishing has led to insecurity -Pollution overflows (due to design flaws) leads to pressure	-Negative impacts on the budget -Routine works (unblocking) -	-Civic education	-Enforcement

	on the workforce.			
<i>Lack of a landfill</i>	-Street scavengers	-Pollution, littering -Pollution and compliance	-Plan in place to construct a landfill (CIP) -By laws exist -Inspectorate unit exists -Improving collection of wastes	-No designated landfill
<i>Raw sewage</i>	-Diseases -	-More work (unblocking) for the workforce -Routine works	-Vaccinations -Soap and milk provided	-Harmful/toxic gases & microorganisms -Health issues
<i>Air pollution</i>	-Air pollution	-Air pollution	-Improving collection	-

c) Wellbeing and equity indicators

The County Government is in charge of waste handling in conjunction with individual waste pickers, Community Based Organisations (CBOs), and other private sector companies. While the County Government and private enterprises normally handle bulk/primary waste from the transfer stations or large properties (e.g. shopping centres, hotels), CBOs and individual waste pickers collect wastes from door to door. The cost of waste collection is borne by the residents, who spend between Kshs. 200 and Kshs. 2,000 per month per property. The waste collected is then dumped at official waste transfer stations with only limited quantities being recycled (NEMA, 2016).

Mombasa County has 72 transfer stations; 32 stations within the Island, 21 in Mainland North, 9 in Mainland South and 10 in Mainland West. Of these 72 stations, 65 are trucks while 7 are formally designated waste transfer spots. The transfer stations do not adequately serve settled areas on the periphery especially in Mainland North and Mainland West. Unfortunately, considerable distances between formal waste collection points and residential areas leads to dumping or indiscriminate burning of waste. High income areas located at a distance of 1km from formal collection points such Mainland North are well served by door-to-door waste collection services which takes waste directly to the Mwakirunge dumpsite (NEMA, 2016).

Table XX. Distribution of Transfer Stations within the County

Area	Population (2009 Census)	No. of Collection Points	Population/Collection Points Ratio
Mainland North	380,055	21	1:18098
Mainland South	166,008	9	1:18446
Mainland West	250,179	10	1:25018
Island	143,128	32	3:147917
Total	939,370	72	1:13047

Source: Field Survey, 2014 (NEMA)

Table XX: Population and estimated waste generation rate for Mombasa County (Palfreman and Clark, 2015)

Socio-economic type	%	Population (2009)	Population (2015)	Per Capita (kg/day)	Waste Production/ Tonnes (2015)
Low	53.6	503,446	590,343	0.55	324.7 (37%)
Medium	23.9	224,613	302,458	0.89	269.2 (31%)
High	22.5	211,311	211,199	1.33	280.9 (37%)
Total	100	939,370	1,104,000	0.80*	875 (100%)

It is clear that the respondents were uncertain about the ratio of male to female employees in the sector as there was no available data from official sources, and between them they gave a range of >20% to <80%. Similarly, there was no reliable figure on the percentage of people employed in the sector at County level relative to the total workforce within the County. They generally agreed that amongst the low income earners, work income is not the sole livelihood for the employee household, and that the contribution of the sector to the household budget ranges from very low to medium.

d) Green economy policy interventions

The respondents provided information on a number of GE initiatives that relate to the waste sector, and outlined the challenges and opportunities as described below.

Mombasa County loses 548 million KES each year due to poor sanitation, and this includes losses due to access time, premature death, health care costs and productivity (Ministry of Health, 2014). This estimate does not include some costs that could be significant (such as water pollution and tourism) and is therefore likely to under-estimate the true cost of poor sanitation (Ministry of Health, 2014).

GE Initiatives	Challenges	Opportunities
-Greening the county (planting trees and other lesser forms of plants (grass)) -Civic education on good disposal methods -Use of materials (in the offices) that don't pose danger to natural resources -Waste segregation promotion by developers.	-Vandalism of infrastructure -Resource allocation -Poor attitude -Vandalizing of litter bins -Compliance failure -Technical skill lacking	-Sludge(?) -Revenue from recycling material -Revenue from energy production. -Conversion of sludge to charcoal

-Waste management and environmental policies/regulation for the County by-laws.	-Lack of capacity -Sludge produced is not fit for use (heavy metals) -Perception of sludge is negative	-Organic waste can be used for fertiliser/manure -Fuel production from waste.
---	--	--

Respondents Views (Planners)	
Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	-There are plans to improve the sanitation facilities, given that only a small part of the population is served by conventional sewerage system. -Greening the County - tree and grass planting. -Civic education on good waste disposal methods -Working with developers to promote waste segregation -Enacting waste management and environmental policies/regulations for the County by-laws. -Consultations and active engagement -Planning for budgetary -The environment department is central to decision making due to the primary source of incomes(?) includes tourism (all development projects have environmental considerations).
Key steps in planning process (NB: county planners only)	-Through the top management (M.D). -Submission at start of financial skills -All projects are required to have an EIA -The city is in the process of developing a SEA.
Timelines for planning process	-Dependent on project magnitude -Adhere to the master plan where there are short term, medium and long term schedules. -1 year (Annual Cycle) to 25 years (SEA) span -
Experience in the County development planning process.	-Departmental Development Planning

Key objectives of your own institutions development plan.	-Making sure sewage is collected, transported and disposed of. -Short term: include improving infrastructure. -Long Term: development of the disposal site. -To create a clean conducive environment for investment growth and habitation by the country residents.	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	Master plans Strategic plans Informal procedures
	b) tools	SWOT Cost/ benefit analysis Rapid results initiatives Performance contracts Planning on a need basis Monitoring committees (sectoral). EIA SEA
	c) platforms	Monitoring committees (sectoral) Public participation Sectoral consultations (departmental) Public Private partnerships Community/Stakeholder involvement.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Operations Buying truck, bulldozer, supervision cars Beautification (greening) Reviving island water plant (Kizingo)	
Above activities budgeted for and implemented	Implementation ongoing	
From your experience, how is monitoring and evaluation done for implementation of the development plans	Poor implementation This is an inspectorate department: does self –audits and NEMA audits (for discharge of sewage).	
Your expectations for a toolkit and knowledge management platform	Marine environment to be considered - pollution as a result of effluents discharge into it Makes it easier to interrogate issues and convince decision makers/strategic organisations in the planning process Mainstream environmental issues identified in the CIDP (County Integrated Development Plan).	

2.2. Tourism

Mombasa County hosts several tourist attractions and world heritage sites including, as outlined in the Mombasa County Annual Development Plan (2014): (i) the historic Fort Jesus Museum which is also a UNESCO World Heritage site; (ii) the gigantic “Elephant Tusks” along Moi Avenue; (iii) several buildings in the old town including the Old Port; (iv) The white sandy beaches (v) Mombasa Marine Park, which is home to a variety of fishes and other sea creatures, and (vi) two private nature trails, Haller Park and Butterfly Pavilion, operated by Bamburi Cement factory.

The County Government has defined eco-tourism zones where only passive activities such as research, forest recreation and observation that do not damage the environment are allowed (County Government of Mombasa, 2015a). The industry has faced strong competition from other destinations, and is hampered by poor tourism infrastructure, lack of capital to improve infrastructure, population pressure on land, advisory notices, and security issues in general (Kenya Water Report, 2005) and the situation is today still pertinent to Mombasa County. The recent security challenges have greatly affected the tourism sector leading to low tourist arrivals, general loss of jobs and negative multiplier effect on other sectors of the County economy like trade and investments (County Government of Mombasa, 2015b).

In general, while there exists a lot of data relevant to the sector at the national level, there was very little that has been done to generate data at the County level (see Annex XX - Mombasa County Data Sheets)

a) General economic sector indicators

There wasn't any data available on the GDP or growth of the tourism sector at county level, but at the national level it contributes 11.6% to the country's GDP. The respondents, however, felt that the sector contribution to the County GDP and employment was, at the very least, important. The occupancy rate of hotel beds in 2013 was 64.4 % and the number of tourism enterprises was 201 (Mombasa County Annual Development Plan, 2014). Gross and per capita emissions of CO₂ from the sector were generally unavailable.

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Important to very important
Economic significance - Employment	Quite important to very important
Key sources of energy for the sector	HEP, KPLC lighting/ electricity from national grid for housing, solar, LPG, gas, charcoal for

	cooking, Hydropower, fuel (petroleum), firewood, diesel.
--	--

Dilapidated road and water transport (e.g. ferry) infrastructure has made Mombasa County unattractive to tourists: conference tourism, marine-based tourism, hotel investment, and eco-tourism are proposed to help spur the growth of the sector (Owinga and Datche, 2015).

b) Environmental issues and related indicators

Environmental related indicators for the tourism sector as per the data sheets (Annex XX) were not available, even though there have been many environmental studies carried out, and specific quantities evaluated either for specific firms, areas, or sub-sectors at scales lower than county level.

Poor management of waste in the waste sector has the highest impact of all sectors on the tourism sector. Solid and liquid effluent discharge from industry, municipality, hotels, residential areas and others all have major negative impacts on the terrestrial and marine ecosystems of Mombasa. Addressing this issue and land degradation will help to greatly reduce the factors that negatively impact on the growth of the tourism sector. Youth activities are concentrated in tourism and recycling (scavenging for food and items that can be resold) (Hildebrand and Holst, 2015) and hence can play a great role in green economy issues within the county. Tourism is climate sensitive because climate dictates the type of natural attractions and resources like the sun and ecosystems, but also determines the nature of tourist activities, their location and the seasons to be spent by the tourists (Njoroge, 2015). Coastal towns such as Mombasa are particularly under threat of sea level rise (cf. Awuor et al., 2008; Ongoma and Onyango, 2014). In Mombasa County, it was observed that the tourism managers, even though having experienced some impacts of climate change such as water scarcity, coral reef bleaching, changes in rainfall, flooding, changes in beach structure, etc., they seem not to have been able to link these local impacts with the global picture that they have of climate change (Njoroge, 2015). On the other hand, government officials, conservation and research institutes were more aware of climate change potential impacts at local scale: thus, while these stakeholder groups would plan for adaptation, the tourism managers would not have tangible adaptation plans (Njoroge, 2015).

Respondents (Planners)	Views	
Indicators		Remarks
Renewable energy used		Solar, Biogas (very low)
Resource efficiency - energy use		Not efficient to highly efficient

Resource efficiency - products/materials recycled	Waste water (Severin hotel), biomass, waste water/sewage, building waste, plastic bottles, linen, building poles.
Waste water treatment and management methods	Water recycling (Severin), septic tanks, cess pits, treatment plant channelling to gardens/irrigation, biodigester, environmentally friendly chemicals.
Ecolabelling of products from the sector	Lighting system bulbs and KAM on energy, eco-jiko, coco-briquettes, environmentally friendly detergents, pest chemicals, ecotourism, waste separation.

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Poor sewerage/waste management</i>	-Increase in effluent discharge to the ocean -Lack of knowledge on proper waste disposal -Attitudes towards proper disposal -Treatment of diseases related to waste control e.g. typhoid. -Buildings collapsing	-Polluted beach and ocean -Eyesore to tourists and local people -Possibility of disease outbreak -Extra cost to the operation -	-Waste management and recycling -Bins -Waste disposal licenses -Health benefits issued to staff	-Lack of coordinated efforts -Lack of infrastructure -Lack of awareness of environmental stewardship -Lack of technical and human capacity -Financial constraints for conservation education and training programmes. -No land for disposal and recycling plant. -Follow up of environmental audits. -Cost of putting up plants and machinery very high. -Staff turnover
<i>Global warming</i>	-Cost of running hotels is higher because of the power bills	-Expense on power bills -Intense warm weather making it unbearable to work in good conditions	-Tree planting -Proper planning and building of houses to enable air circulation in town. -Solar and wind energy	-Financing on training and greening initiatives at school level
<i>Environmental degradation</i>	-Lack of knowledge and mitigation to	-Compromise on standards needed in	-SOP refreshers	-Maintaining and monitoring the standards is difficult. -High cost

	environmental issues	Environmental management		
--	-------------------------	-----------------------------	--	--

c) Wellbeing and equity indicators

It is noted that tourism planning within a region provides for socio-economic development, environmental protection and conservation (Mombasa County Government, 2015). Employment in the tourism sector is estimated at 37,535 people, based on the fact that 4% of the total County population of 939,370 is employed in the sector according to the national population census of 2009. The respondents estimated the ratio of male to female employees in the sector to be between 40 and 60 per cent. For some, the work income is the sole livelihood, but others have small scale income generating activities to supplement their earnings. The sector contribution household budget ranges from low to very high. The estimated percentage of people employed in the sector at county level relative to the total workforce ranged from 41 to 80 per cent according to the respondents.

Msengeti and Obwogi (2015) carried out a study on employees in the hotel industry in Mombasa County, and concluded that pay as a financial component of reward has a weak influence on employee retention in hotels in Mombasa County tourism area as compared to the non-financial element of work environment. It was further noted that there were also significant pay discrepancies between the players in the industry leading to employees constantly seeking better opportunities, further exacerbating the low employee retention issue (Msengeti and Obwogi, 2015).

d) Green economy policy interventions

The development and success of hotel industry, for example, depends on the availability of natural resources since the latter are one of the main assets of the local hotel industry, but on the other hand, hotels are among some of the largest consumers of energy, water resources and consequently generate a large amount of waste (Gitobu and Njoroge, 2015). It has been shown, from a survey of 24 hotels in Mombasa County, that environmental conservation, profit increment and market advantage has to a very large extent driven hotels in Mombasa County, Kenya to adopt green marketing practices such as; product modifications, changes in the production process, elaborate advertising, change in packaging, and selling of products based on environmental benefits (Gitobu and Njoroge, 2015). Mombasa County should strive, as a matter of priority, to ensure that all tourism market players in the County have access to the requisite finances needed so as to effectively compete with tourism players from other Counties (Owinga and Datche, 2015).

GE Initiatives	Challenges	Opportunities
-Coast Region Green Growth Initiative	-No corporate social responsibility to support GE growth as it is a non-profit making body -No formal monitoring and evaluation system -More funding is required -	-Advocacy and collaboration projects -
-Tree planting -Sensitization of community green economy -Clean up events -Green school project	-Funding: Most of the schools/ institutional based GE need funding - so as to undertake trainings to teachers and students -Space: in most town schools there is little space to undertake GE initiatives. -	-The community welcomes the initiatives -Most schools and teachers and management of schools have keen interest and are willing -Proposals for support e.g. for tree nursery, planting, beautification, botanical and institutional gardens. -
-Kenya Coastal Climate Change Adaptation Project -	-	-
-Coastal Water Security and Climate Resilience Project -	-	-
-Solar lamps for street lighting. -Construction of green building with a mega solar P.V. system to provide services instead of electricity.	-Solar system - initial cost too high -	-Payback period is short - can be extended even to power lighting -
-Water waste treatment plant	-Water purification very expensive	-WCK as an institution has staff and volunteers willing to undertake these initiatives. -Bottling of fresh water can be done in-house and same sold to clients of other institutions.

Respondents Views (Planners)	
Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	Employees provide information that are relevant to County planning.
Key steps in planning process (NB: county planners only)	Not stated
Timelines for planning process (NB: county planners only)	Not stated
Your experience in the County development planning process.	Environment Department

Key objectives of your own institutions development plan.	To achieve the standards expected internationally and locally.	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	Collect real data from institutions
	b) tools	Audits and Issuing of permits to those who comply
	c) platforms	Seminar forums and through umbrella authorities.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Construction of Green Building Staff Training and Development	
Above activities budgeted for and implemented	Staff Training and Development has been implemented	
From your experience, how is monitoring and evaluation done for implementation of the development plans	Putting up water and power meters to give data on real capacity consumption through inspections.	
Your expectations for a toolkit and knowledge management platform	Expectation is to analyze current status and project goals to be achieved by a certain period using available and borrowed resources.	

2.3. Cities and Built Environment

Land ownership in most areas is not guaranteed as most of the residents do not legally own land and the land they live on is owned by absentee landlords (Mombasa County Annual Development Plan, 2014). A number of informal settlements exist in the County, and the growing population creates a huge demand for quality and affordable housing (Mombasa County Annual Development Plan, 2014).

a) General economic sector indicators

There has been an increase in real estate investment with the construction of beachfront apartments, hotels and resorts concentrated along the low-lying areas of the shoreline (Government of Kenya, 2010). These infrastructural developments increase shoreline instability and imperil the stability of seawalls or buildings due to high-energy waves from intensive erosion (Odada, 1995; Mwanje, 1997; Kairu and Nyandwi, 2000).

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Quite important to very important
Economic significance - Employment	Important to very important
Key sources of energy for the sector	Electricity, hydropower and fuel

b) Environmental issues and related indicators

Coastal erosion has been exacerbated by human occupation and intensive development of infrastructure that is concentrated along the coastal strip (Omuombo et al., 2013a). The tourist hotels and resorts keep their beaches clean through the clearing of seaweed, and creation of beach gaming and relaxation areas has increased the human traffic in the beach, thus exposing the loose sands that are easily washed into the sea during the high tides (Omuombo et al., 2013a), leading to the characterization of these areas as coastal erosion ‘hotspots’ (Government of Kenya, 2009, 2010). In Diani–Bamburi area, shoreline recession of up to 10 m in 10 years is reported to have occurred over the period 1996–2006 (Ballot et al., 2006) due to modification to current flows caused by the construction of seawalls to protect the beachfront tourist hotels and resorts. The ecological conditions are evolving fast due to numerous developments in the County, including the recent dredging to deepen the Kilindini Channel of the port of Mombasa (Mombasa County Annual Development Plan, 2014)

Frequent storms are expected off the East African coast due to climate change-related global sea level rise (Omuombo et al., 2013a), and it is estimated that by 2100, 17% of Mombasa city could be submerged by a 30 cm sea level rise (Awuor et al., 2008; Kabede et al., 2009). As a consequence of these changes, the low lying areas of Mombasa County, and the Kenya coast in general is at risk of related adverse impacts to biodiversity, land, properties, and the disruption of economic activities beyond the country’s borders since it has East Africa’s largest seaport (Kabede et al., 2009; Brown et al., 2011).

Pollution from the export processing zone, industries and mining has not been quantified, although these activities are known to contribute to pollution of the coastal waters and ecosystems as the industries discharge untreated solid and liquid effluents into the sea (Government of Kenya, 2009, 2010; Hashim et al., 2004), as well as soil, vegetation, and groundwater pollution as solid and liquid effluents are discharged on land (cf. Owuor, 2011). High concentrations of Co, Cr, Cu, Ni, Pb, V and Zn have been measured in the surface samples of sediments from Makupa Creek, Mombasa City, and have been attributed to urban development in the industrial city (Williams et al., 1997; Muohi et al., 2003). Quality poles have been wiped out in most mangrove areas of Mombasa and this is associated with high human populations (Abuodha and Kairo, 2001).

Respondents Views (Planners)	
Indicators	Remarks
Renewable energy used	Solar, Biogas, Hydro, Electricity
Resource efficiency - energy use	Not efficient to slightly efficient

	Moderate Old Architecture- less cooling required - not very efficient.
Resource efficiency - products/materials recycled	Plastic and paper Iron sheets (hoarding), scaffolding, cement bags
Waste water treatment and management methods	Kizingo & Kiparu treatment Septic tanks, soak pits, municipal sewage. Biodigester/septic tank, soak pits
Ecolabelling of products from the sector	Solar, eco-jiko, coconut briquettes Paints

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Land disturbance</i>	-Poverty	-Displacement	-Civic education -Backfilling	-Poverty
<i>Deforestation</i>	-Due to ignorance	-Low water supply especially in cases of water pans	-Encourage tree planting	-
<i>Waste Management</i>	-Collapsing of buildings -Industrial discharge	-Poor development -Hazards	-None	-Lack of capacity -Community not sensitised.
<i>Poor Sewerage</i>	-	-	-	-Public awareness -Flooding in old town -Drainage -
<i>Environmental Degradation</i>	-Cutting down trees	-	-Planting trees -Sensitisation programs -Compensation and engaged on contractual basis.	-
<i>Socio-economic changes</i>	-Killing of wild animals e.g. lions -HIV/AIDS	-	-	-
<i>Skills transfer</i>	-	-	-	-

c) Wellbeing and equity indicators

In this sector the ratio of male to female employees is estimated by respondents to be 70% to 30%. The contribution of the sector to household budget is medium to high, and the employee engages in other livelihood activities. The percentage of people employed in the sector at county level relative to the total workforce is estimated by respondents to be between 21 and 40%.

d) Green economy policy interventions

The management of the Kenyan coast is sectoral with the formulation of legislation to regulate fisheries, water, tourism, forests and biodiversity conservation through various Acts of Parliament (land planning, tourism industry, tourism industry licensing, tourism development and cooperation, wildlife conservation and water) (Omuombo et al., 2013b). These acts focus on the management and conservation of the various resources and play a role in the management of the coastal resources. The Environmental Management and Co-ordination Act (EMCA) of 1999 provides a legal and institutional framework for integrated coastal zone management (ICZM) that can bring together all the sectoral management frameworks (Omuombo et al., 2013b). The shoreline management strategy is the basis for the formulation of the National Coastal Zone Management Plan. In this approach, issues affecting predefined individual sediment cells are identified and prioritized as areas of concern with regard to management (Government of Kenya, 2009, 2010; Muthiga, 2009). The success in the implementation of the marine protected areas as a tool for protection and management of ecosystems in the coastal zone has influenced the shoreline management strategy that forms the basis of ICZM to use a site-specific approach (Omuombo et al., 2013b).

GE Initiatives	Challenges	Opportunities
-Delegated to WRMA -Planting trees -Rehabilitation of degraded -Green schools project -Climate change adaptation project -Coastal water security -Climate resilience project -Clean up events -	-High Cost of power -Funding -Conflict of interest in implementation -Incitement, politicisation of project implementation. -Late remittance of funds -Middle cadre is lacking (?)	-Public support and willingness to participate. -Resources available (funds).

Respondents Views (Planners)

Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	Consultative process Representation in county board	
Key steps in planning process (NB: county planners only)	No procedure for engagement	
Timelines for planning process (NB: county planners only)	None 12 months	
Your experience in the County development planning process.	Positive, there is no full rep. (?)	
Key objectives of your own institutions development plan.	Regulation Development: target to raise sum in a year Maintenance and equipment - review equipment Engage locals Pay allowance	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	Strategic plans Performance contract Rudimentary data Techniques mechanics - collect and keep data
	b) tools	EIA log books work tickets Equipment utilisation
	c) platforms	P.P Workshop with water service providers All activities (monthly) for all equipment and consumption - formula used to calculate cost effectiveness.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Infrastructure increased in water service areas Water project in marginalised areas Rehabilitating Baricho treatment works Expanding sewage works	
Above activities budgeted for and implemented	Ongoing implementation process Maintenance of log books for each equipment	
From your experience, how is monitoring and evaluation done for implementation of the development plans	Building inspections - County Strategic plan - departmental objectives	
Your expectations for a toolkit and knowledge management platform	GIS based data base.	

2.4. Transport and Infrastructure

The transport and related infrastructure situation is as follows (Mombasa County Annual Development Plan, 2014):

Roads: There are a total of 257.17 km of bitumen surface roads, 127 km of gravel surface roads and 91.29 km of earth surface roads in the County (Mombasa County Annual Development Plan, 2014). While the major roads are in a fair condition, access roads within the residential and industrial areas are in deplorable state. The roads are maintained by the national government through Kenya Rural Roads Authority (KeRRA), Kenya Urban Roads Authority (KURA), the Kenya National Highways Authority (KeNHA) and Mombasa County.

Railway: The County has ten kilometres of railway line and three railway stations. The County Government of Mombasa envisions having light commuter rail within the city in the long run. Construction of the Standard Gauge Railway Line from Mombasa to Nairobi is expected to start soon.

Water Transport: Mombasa's port is the largest in Kenya, with 19 deep water berths, two additional berths nearing completion and two oil terminals. Rail connects the port to the interior. There is little or no scheduled passenger service. International cruise ships frequent the port.

Ferry: There is no bridge between Mombasa Island and the south coast, instead the distance is served by ferries operated by the Kenya Ferry Service from Kilindini and Mtongwe to Likoni in the south coast of Mombasa. As a result of the major build-up of more luxurious hotels on the South Coast and a lack of a direct bridge linking the South Coast to the North Coast, visiting tourists have the option of flying directly into the South Coast airstrip using the domestic airline, Air Kenya.

Port: The port of Mombasa is also a key resource and the gateway to the East and Central African region, as it serves the entire region's export and import needs. In 2012, dredging was undertaken with the idea of deepening the Likoni channel to facilitate usage of the port by larger vessels. The ongoing construction of a second container terminal at Kipevu will increase the capacity of the containers being handled to three times the current capacity.

Airports: The County has one international airport, the Moi International Airport within Changamwe sub-County. The airport is the second largest airport in Kenya and is used by both domestic and international flights. The airport is essential in the promotion of tourism and investment opportunities in the County and in the Coast region.

a) General economic sector indicators

At the national level, the transport sector contributes about 16% to the GDP, and registered a growth rate of 6 per cent in 2013 (KNBS, 2014). The length of the road infrastructure is 475 km (Mombasa County Annual Development Plan, 2013), but there are no specific data on the population of cars/trucks within the County. Other information for which no data was available at County level are: road and rail freight volumes, total person kilometers travelled, and total public transport boarding per capita. In terms of investment in road infrastructure, the County is embarking on expanding the existing road network in order to make transport affordable and faster for economic efficiency. The Government of Mombasa has contracted firms to improve and construct 14.1 km of roads and installation of traffic lights at six junctions at a total cost of KSh226 million. The works are at various stages of execution and are scheduled to be completed

by June 30 2015 while Aldina-Kwangombe, which includes construction of a drift in the ocean will be completed in the first quarter of the financial year 2015/2016 (Hazel J Koitaba, County Government of Mombasa, Fiscal Strategic Plan 2015). The works to improve and construct 15.13 km of roads and one bridge at a total cost of KSh 292 million have just commenced and were scheduled to be completed by June 30 2015 while others will be completed in the first quarter of 2015/2016.

CO₂ emissions from transport (million metric tons) in Kenya¹ was last measured at 4.71 in 2011, according to the World Bank: CO₂ emissions from transport contains emissions from the combustion of fuel for all transport activity; regardless of the sector; except for international marine bunkers and international aviation. Data is not available at County level for emissions, consumption, NO₂ contributions, as well as costs of damages related to climate variability.

The respondents ranked the economic significance of the sector with respect to GDP as quite important to very important, and with respect to employment as slightly important to quite important.

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Quite Important to very important
Economic significance - Employment	Slightly important to Quite Important
Key sources of energy for the sector	Geothermal, hydrocarbons, Electricity, Solar, fuels, gasoline, diesel.

b) Environmental issues and related indicators

Noise, dust, poor garbage disposal, exhaust fumes, tyre traction and fuel as well as other transported goods spills that lead to environmental pollution, degradation and erosion, are some of the environmental issues related to the transport industry. However, there is no data at County level as per the indicators specified in the data sheet for this sector. Flash flooding as a consequence of storm drainage systems being blocked by solid waste is a problem made worse by the fact that the system itself is largely dilapidated. Oil spill events have generated devastating effects on the ecosystems; in 1988, for example, about 10 ha of mangrove forests was lost, while in 2005, an oil spill from Kipevu oil terminal affected 234 ha of mangroves at Port Reitz (Abuodha and Kairo, 2001; Kairo et al., 2005). The full impacts of oil pollution are still unknown, although it is common to sight discharges from ships and tankers that are washed offshore (Government of Kenya, 2009).

¹ <http://www.tradingeconomics.com/kenya/co2-emissions-from-transport-million-metric-tons-wb-data.html>

Respondents (Planners)	Views	
Indicators		Remarks
Renewable energy used		Electricity Solar power for street lighting.
Resource efficiency - energy use		Efficient
Resource efficiency - products/materials recycled	-	Lami on destroyed roads is used to make pedestrian pathways Used oil - lubrication Milled Asphalt layers are blended with gravel to give a superior base quality material Topsoil that is stripped is used again on the side slopes for grassing Rock Used paper - for making tissue paper Effluent tests done every 2 months - before this is released to the municipality.
Waste water treatment and management methods		Channelled to the drains Waste water is treated through a plant where contaminants are removed The semi-solid waste (sewage sludge) also undergoes further treatments before being disposed/applied to other uses like land application.
Ecolabelling of products from the sector		-

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Noise (Office?)</i>	-Community Complaints	-	-	-
<i>Dusty roads/ Air Pollution</i>	-Community Complaints	-	-Come up with cleaning machines with H2O -Environmental measurement done for air, soil and water every year	-
<i>Poor garbage disposal/ Polythene bags</i>	-	-	-	-
<i>Dust and fumes from rock-crushing plants and asphalt mixing plants</i>	-Communities living around areas with rock crushing and asphalt mixing	-Delayed delivery of projects	-Dust/fumes control mechanisms	-Sometimes dust/fumes levels are high when there is a

	plants may protest and stop works			breakdown in the dust control systems
<i>Exhaust fumes from motorised traffic</i>	-Air Pollution	-Air Pollution	-In the long run, types of fuels need to be eco-friendly	-Alternative sources of fuels to run motorised traffic not researched/not in use.
<i>Soil erosion emanating from change in land use.</i>	-May lead to displacement of people -May lead to collapse of embankments	-Washout/undermining embankments	-Protection works to be done e.g. gabions	-Land use keeps changing and therefore erosion is almost inevitable.
<i>Environmental degradation</i>	-Spillage	-	-Immediate action in case of spillage	-Capital intensive -Delays due to limited capacity/service providers.
<i>Contamination of water sources</i>	-Water contamination	-Waferina community in Kibwezi already contaminated.	-Monitoring systems in place -Treat the contaminated ground.	-
<i>Sink Holes</i>	-Soil degradation/erosion	-	-A consultancy has been engaged to suggest relevant measures	-
<i>Noise from aircrafts / traffic</i>	-	-Inaccessibility to a nearby hospital due to heavy trucks	-	-
<i>Emissions /Fumes</i>	-	-	-	-

c) Wellbeing and equity indicators

The number of people employed in the sector at County level is not specified. The Dongo-Kundu by-pass is expected to ease congestion at the Central Business District, as traffic from Nairobi to the South coast shall be diverted at Miritini towards Likoni and Diani. The County Government

of Mombasa is also planning to have a second bridge fixed to the north mainland to ease traffic congestion to the north.

The County Government has procured road construction equipment for in house road construction and maintenance. These include two motor graders, two wheel loaders, two drum rollers, one backhole excavator and five tippers this financial year. These are intended to increase coverage in opening of access roads while reducing the cost of works compared to contracting.

This is aimed at a continued focus on road infrastructure improvement through maintaining the already existing road networks while also opening up new access roads so as to restore confidence of residents and making the cost of doing business favorable. Transport infrastructure development will also help revamp the tourism sector and foster social integration in the County.

d) Green economy policy interventions

GE Initiatives	Challenges	Opportunities
-Collaboration with KFS to plant trees along roads -Waste management initiative - put bins around beaches, grassing such areas -Tree planting in project site - Kaloleni -Grass planting (scour checks) - on slopes to check soil erosion. -Compensation of forest/vegetation cover after bush clearance is carried out - through afforestation - a case in point is at the Ngong road interchange with the Southern bypass where KENHA staff did afforestation on the adjoining land. -Planting trees around the office -Sensitisation of staff on power conservation -Environmental management system audit -Target of zero waste on hydrocarbons - monitor system and	-Some trees can't grow well in town centre -Limited space especially in town centres -Costly -Tree planting after site clearance has to wait until the pavement is constructed. This may bring about disquiet among communities, conservation (environmental) lobby groups etc. -Lack of capacity -Budgetary constraints - high initial investments -Lack of sufficient policies -Climatic effects -drying of trees -Lack of awareness/resistance	-Community willing. -The opportunity offered is that forest cover is not only reinstated but also in some instances the cover is extended/intensified more than the original cover. -Policy guidelines - could be an alternative -Regulations - in place for use of alternative sources of energy. -KPC being a nationwide institution gives it a better position for implementation on GE initiatives -Partner with other institutions.

ensure no leakage -spillages are promptly attended to. -		
---	--	--

Respondents Views (Planners)	
Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	Share plans with county guys (?)
Key steps in planning process (NB: county planners only)	Community Involvement Other stakeholders involvement Involved the consultant and stakeholder engagements Drafting Sensitise community and social audiences
Timelines for planning process (NB: county planners only)	5 years
Your experience in the County development planning process.	Community Involvement - planning Overlapping of projects with the country Participative - quite good N/A
Key objectives of your own institutions development plan.	Strategic Takes into account all challenges experienced Risk management Development of pipeline to meet demands in the sector.
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies Stakeholders participation Community engagement - planning, tender evaluation Proper strategies in place Enterprise risk management system
	b) tools Laboratory testing of materials Wi-Fi for customers - budgeting Good working environment Monitoring and evaluation
	c) platforms Forums for different airports/stakeholders Consultations Performance contracts Sensitisation
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Changamwe road, Roads planned for - concrete block payments/ access roads - in Changamwe, Jomvu, Nyali, Kisauni, Kisauni, and Likoni constituency. Water supply system - a consultant engaged by World Bank Power supply and distribution consultant - to look on energy saving and green options Pipeline from Mombasa to Nairobi - started August last year Social responsibility - dispensary at Makindu
Above activities budgeted for and implemented	Not yet Some solar panels already installed in Lamu.
From your experience, how is monitoring and evaluation done for implementation of the development plans	Monitoring and evaluation committee, community evaluation. Not really taken into consideration for GE Monitoring and evaluation reports Controls monitoring

2.5. Fisheries

Mombasa County boasts of fisheries as an important resource that consists of artisanal and pelagic fisheries. Traditionally, inhabitants of the County have relied on fisheries and mangrove exploitation. Unpublished statistics by Fisheries Department, estimate that over 2000 fishermen are directly engaged in artisanal fishing with most fishing activity undertaken along the reef (NEMA, 2013). The fishermen's main fishing gears include hook and line, gillnets, shark nets, beach seine and basket traps (malema). They also use simple fishing vessels (mainly dugout canoes) and gears. Types of fisheries include marine and coastal, dam and reverie, aquaculture (tilapia and mud fish). Fish farming practices in Mombasa County include pond culture, raceways culture, and tank culture.

a) General economic sector indicators

The marine fisheries resources are both inshore and offshore: the inshore fish species are overexploited by local fishers, while the offshore resources outside the territorial waters are exploited mainly by Distant Waters Fishing Nations (DWFN) (FAO, 2007). The marine fisheries contribute on average only 4 per cent of the total national fish landings (FAO, 2007). There are no County level data for the suite of general economic indicators as specified in the data sheets (Annex XX). However, extraction of fish and other resources from the narrow continental shelf, coral reef and mangrove ecosystems increases each year with inadequate monitoring and management structures to protect the resource bases. Mombasa has 19% of the fish landing sites, with the other towns along the coast having the following: Kwale (29%), Kilifi (18%), Malindi (19%), Lamu (12%), and Tana Delta, (3%) (Ministry of Fisheries Development, 2012). The respondents ranked the economic significance of fisheries in terms of GDP as quite important to very important, and in terms of employment as being slightly to quite important. Various energy types are used in the sector as indicated below for transportation, freezing and processing of fisheries products.

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Quite important to very important
Economic significance - Employment	Slightly important to Quite important

Key sources of energy for the sector	Electricity, generator, charcoal, fossil fuel, wood fuel, diesel.
--------------------------------------	---

b) Environmental issues and related indicators

Serious threats to fisheries in the area come from the ever increasing human pressure through tourism, industrial pollution, over fishing, destructive fishing, mangrove logging and other unsustainable uses (cf. Kairu et al., 2001). Experience over the years has proved the value of gazettement, such as that of the Mombasa Marine Reserve, as a primary conservation tool even if resources often lack to follow through on monitoring and active management (Githitho 2004). Anthropogenic activities, particularly crop production alongside the discharge of raw domestic and industrial effluents coupled up with agrochemical wastes have negatively impacted upon the floral and faunal biodiversity within the Tudor Creek ecosystem, particularly impacting the fishing industry that supports a large population of the ecosystem's human population (Government of Kenya, 2009). In 2006, Beach Management Units (BMUs) were implemented through the Ministry of Fisheries in a co-management approach and to date they are still doing so. The BMUs empower the local communities as decision-makers in the management of the resources and thus work with the government to enforce legislation on fishing and conservation (McClanahan, 1997; Cinner et al., 2012).

Table XX. Fisheries production systems and their impacts to the environment

Production system	Impacts	Mitigation measures
Marine and Coastal fisheries	Introduction of biodegradable substances e.g. oils & detergents	Enactment and enforcement of existing laws Limiting of access to the fishery Awareness creation to resource users
Aquaculture	Introduction of effluents from agriculture inputs into the water system Soil erosion and silting	Regulation of pond construction Regulation of types of inputs Soil erosion control practices

Source: Draft MSUDP

Respondents Views (Planners)	
-------------------------------------	--

Indicators	Remarks
Renewable energy used	Energy-saving jikos, fireless cooking, firewood, electricity, charcoal, briquettes, solar
Resource efficiency - energy use	Not Efficient to Efficient
Resource efficiency - products/materials recycled	Waste paper Farm produce Manure/fertiliser/compost
Waste water treatment and management methods	Harvesting Purifying County sewage infrastructure Sometimes untreated
Ecolabelling of products from the sector	N/A

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Water Shortage</i>	-High cost of inputs -Pests -Knowledge is low	-	-	-Limited funds to support extension services. -
<i>Drought</i>	-Food shortage -	-Loss of livelihood crops -Discouragement and change of priority	-Subsidy inputs	-Overreliance on subsidies
<i>Flooding</i>	-Diseases	-	-Joint planning	-
<i>Harsh Climatic conditions</i>	-Poverty	-Mangroves act as a breeding place for fish	-Sensitise the communities on the importance of mangroves	-
<i>Deforestation</i>	-Poverty	- Planting more trees -Land rehabilitation	-	-Poor attitude to afforestation
<i>Air pollution</i>	-Health problems e.g. lead pollution	-	-	-Lack of co-operation from the industrial sector
<i>Oil Spillages</i>	-Rare but heavy pollution impact	-Death of marine life e.g. fish	-	-
<i>Sewage management</i>	-Disease	-	-	-

Waste Management	-Illegal dumping -Sand harvesting	-Death of marine life e.g. fish -Bureaucracy	-More deep sea fishing boats -Compensation of fishermen	-
Emissions	-	-	-	-

c) Wellbeing and equity indicators

The number of people employed in the sector is not specified at the County level, but the number of fishers in Mombasa is about 1500 (Ministry of Fisheries Development, 2012). The respondents indicated that the ration of male to female employees in the sector is 9:1, and that for some, work income is the sole livelihood for the employee household. The contribution of the sector to household budgets was ranked as medium to very high. There was a large uncertainty about the percentage of people employed in the sector at the county level relative to the total workforce, ranging from <20 % up to about 80 %.

d) Green economy policy interventions

The Ministry of Fisheries Development in recognition of depleted wild fish stocks in both fresh and marine water, and to meet the objective of promoting aquaculture, begun the Economic Stimulus Programme (ESP) in order to highlight fish farming as a viable economic activity in the country by raising the income of farmers and other stakeholders in the fishing industry (Ogada and Nyingi, 2013).

It was noted that up to 2013, the nation was still awaiting the decision of parliament in regard to the fisheries policy (Ogada and Nyingi, 2013). Today, the Agriculture, Fisheries and Food Authority (Amendment) Bill is still pending and in August 2015 the President called for its speedy passage. Ogada and Nyingi (2013) note that when the Fisheries 2008 Policy is enacted into law, the government will make institutional arrangement to improve coordination of research and management of fisheries including the following: create the Kenya Oceans and Fisheries Council chaired by the minister to advise and review the oceans and fisheries sector; create the Kenya Oceans and Fisheries Service (KOFS) to harmonize fisheries extension, development, management and marketing; and restructure the Kenya Marine and Fisheries Research Institute to offer effective and timely research to guide KOFS.

GE Initiatives	Challenges	Opportunities
-Branding of staff and stakeholders -	-Demoralisation by detractors -	-Formation of groups to do joint bidding -
-Training of fishers/farmers	-Fishing conditions	-Awareness and sensitisations

-	-Knowledge and low absorption -	-Processing of fruit and vegetables
-Tree planting -	-	-Improve forest cover
-Energy saving jikos	-Fireless cookers were viewed as an expensive undertaking. -Metrics for developing those targets and maintaining -	-Alternative jikos -
-Solar energy, wind energy -	-High cost of renewable energy -	-
-Protection of mangroves	-	-

Respondents Views (Planners)	
Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	Committees Planning Water Energy Tourism
Key steps in planning process (NB: county planners only)	N/A Stakeholder consultation Bottom up approach
Timelines for planning process (NB: county planners only)	N/A
Your experience in the County development planning process.	N/A
Key objectives of your own institutions development plan.	N/A Increase tree cover Enhance revenue through sustainable forestry Improve institutional capacity Growth of the sector Develop industry output and capacity
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies Training and capacity building Having yearly target Public/private participation Stakeholder involvement Expert consultation
	b) tools pamphlets Visiting communities to give technical advise Media /Radio for communication
	c) platforms Field days Radio/media Sensitisation through public fora
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Rehabilitation of catchment areas Protection of bamboos seeding production Restoration of degraded forests 38Ha.

	Protection of tree and bamboo growing 47ha.
Above activities budgeted for and implemented	Seeding production by farmers
From your experience, how is monitoring and evaluation done for implementation of the development plans	Field visits to assess success rates Protection of seeds and replanting of dead seeds Monitoring and evaluation for projects quarterly, half-yearly and annually, performance contract, ISO certified, global benchmarking.
Your expectations for a toolkit and knowledge management platform	Effective communication of knowledge. Capacity Building at all levels

2.6. Energy

The Kipevu power plant produces power which is fed into the national grid, and there are plans to construct an 800MW LNG power plant in Dongo Kundu area (Mombasa County Annual Development Plan, 2013). The report also notes that the County has a high potential for generation of solar and wind energy, but this remains unexploited. Kipevu III power station located at Kipevu in Mombasa is the most recent and largest diesel plant in East Africa. Other sources of energy are wood, charcoal, kerosene, LPG and biogas.

a) General economic sector indicators

The respondents noted that energy sector is very important with respect to the county GDP and employment. Ngugi (2013) notes the following: (i) Only nine per cent of residents in Mombasa County use liquefied petroleum gas (LPG), and 39per cent use paraffin; (ii) Six per cent use firewood and 41 per cent use charcoal; (iii) the most common cooking fuels among male headed households are paraffin and charcoal at 40 per cent each while the most common cooking fuels for female headed households is charcoal 45 per cent, (iv) Mvita constituency has the highest level of LPG use in Mombasa County at 23%, almost eight times Likoni constituency, which has the lowest share at 3%, (v) Likoni constituency has the highest level of charcoal use in Mombasa County at 46%, while Nyali constituency has the lowest at 37%, (vi) Changamwe constituency has the highest level of use of paraffin at 49%, and Mvita constituency the lowest at 20%, (vii) Mji wa Kale/Makadara ward has the highest level of electricity use at 97%, which is 95 percentage points above Mwakirunge ward which has the lowest level of electricity use.

Respondents Views (Planners)	
Indicators	Remarks
Economic significance - GDP	Very Important
Economic significance - Employment	Very Important

Key sources of energy for the sector	By-products (crude oil) diesel, emergency power.
--------------------------------------	--

Mombasa County has an electricity supply gap with current demand being around 350MW and is projected to be 1470MW by the year 2030 as shown in the table below:

Current and Projected Power Demand

2015	2017	2020	2025	2030
350MW	400MW	530MW	870MW	1470MW

b) Environmental issues and related indicators

Respondents Views (Planners)	
Indicators	Remarks
Renewable energy used	Geothermal, hydro, wind, solar
Resource efficiency - energy use	Quite efficient
Resource efficiency - products/materials recycled	Oil to heat energy (sludge)
Waste water treatment and management methods	Sold to vendors, segregated and waste disposed.
Ecolabelling of products from the sector	Energy

Environmental Issues/Challenges	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
Prohibitive controls/regulations	-	-	-	-
Poor disaster management system	-	-	-Maintain the system	-
Access is hazardous/pollution	-Smoke emissions	-	-Compliance to World Bank standards	-

c) Wellbeing and equity indicators

The respondents estimated that the ratio of males to females in the sector is 7:3. No information was provided for other indicators.

Mombasa County is well endowed with electricity supply network: the urban area has more coverage than the peri-urban while areas such as Mwakirunge, Maunguja II, DongoKundu and ShikaAdabu lack connections (Mombasa County Annual Development Plan, 2013). It has been noted, however, that the gaps within counties in access to electricity for lighting are also enormous: the most severe disparities in the gap between the ward with the most access to electricity and the least access, more than 40 percentage points, are in Mombasa (95 percentage point gap between highest and lowest ward), Garissa (92 percentage points), and Nakuru (89 percentage points) (Ngugi, 2013). The Tariff arrangement in Mombasa County is similar to everywhere in Kenya. While individual customers pay their bills directly to Kenya Power, the County Government pays for street lighting. The Tariff arrangement has an array of options such as mobile phone, Postal Corporation of Kenya, ATMs among others for making payments.

d) Green economy policy interventions

GE Initiatives	Challenges	Opportunities
-Maintenance of mangroves -Tree planting -Support community -Tree Nurseries	-Finances	-Sustainability

Respondents Views (Planners)		
Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	Not stated	
Key steps in planning process (NB: county planners only)	Not stated	
Timelines for planning process (NB: county planners only)	Not stated	
Your experience in the County development planning process.	Not stated	
Key objectives of your own institutions development plan.	Sustainability Succession planning	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	Strategic Planning
	b) tools	
	c) platforms	Energy sector forum
Activities with a budget allocation in the 2015/2016 development plan for your sector.	Regulatory Framework	

Above activities budgeted for and implemented	Not stated
From your experience, how is monitoring and evaluation done for implementation of the development plans	Not stated
Your expectations for a toolkit and knowledge management platform	A balanced scorecard ISO certification.

3. Prioritisation of Analysed Sectors

3.1. Respondent Analysis

A total of 28 questionnaires were administered in Mombasa County. They were from the priority sectors as agreed by the stakeholders in the inception meeting. The sizes of the institutions varied from small, medium to large enterprises as presented in Figure 7.1-1.

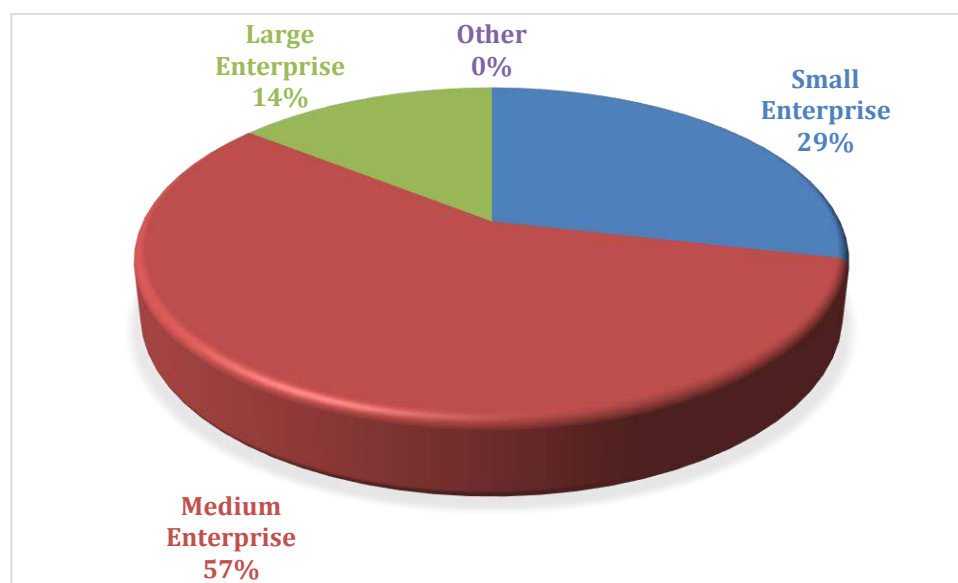


Figure 3.1-1 Institutional Size

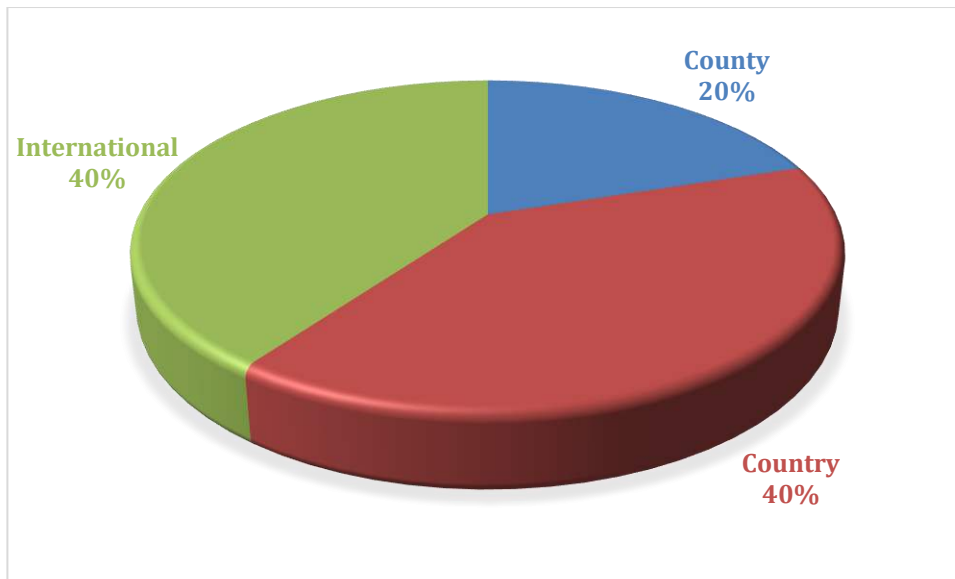


Figure 3.1-2 Institutional Geographical Coverage

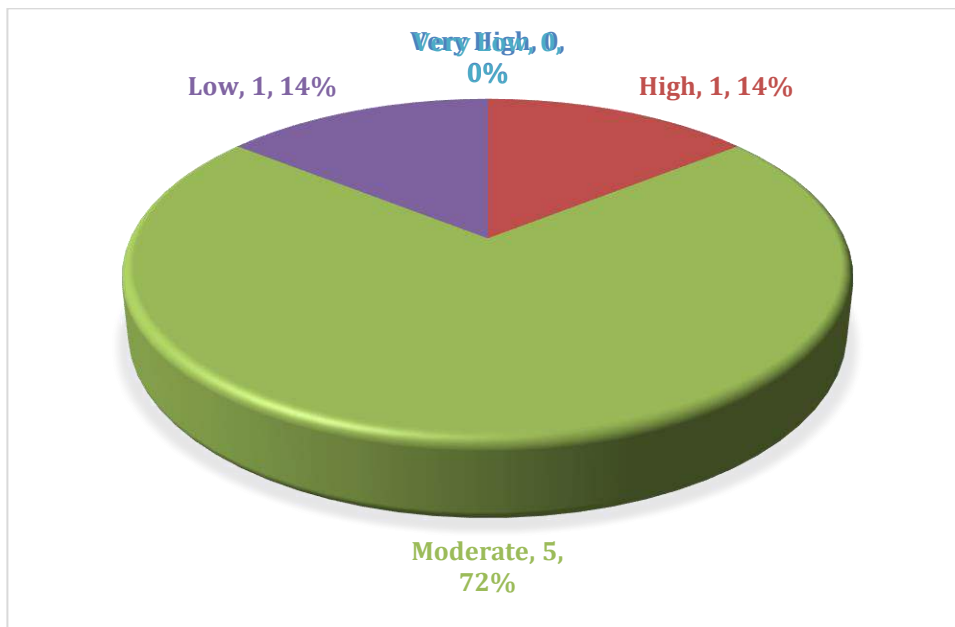


Figure 3.1-3 Institution Market Share

FACTORS CONSIDERED IMPORTANT WHEN IT COMES TO ENERGY SOURCE

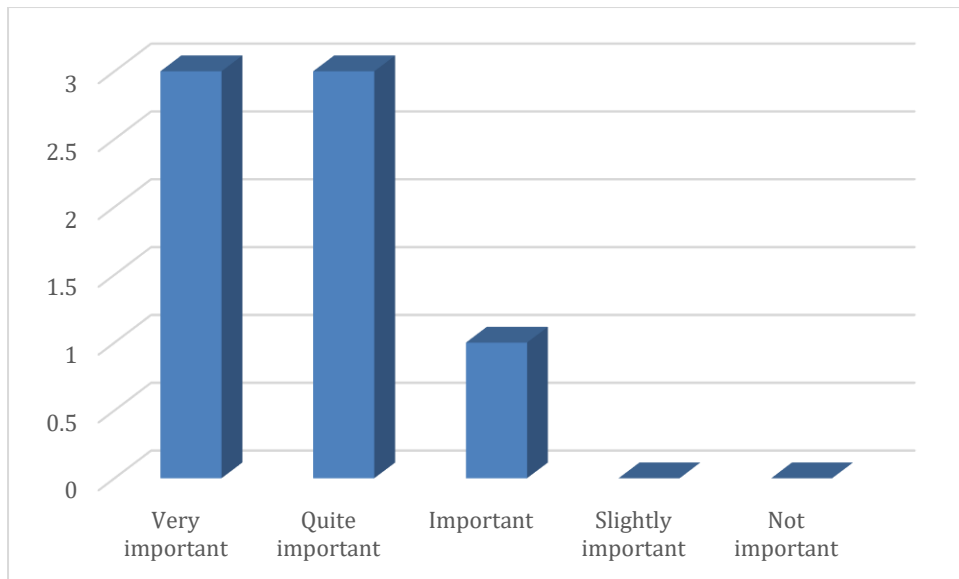


Figure 3.1-4 Reliability as an important factor in energy source

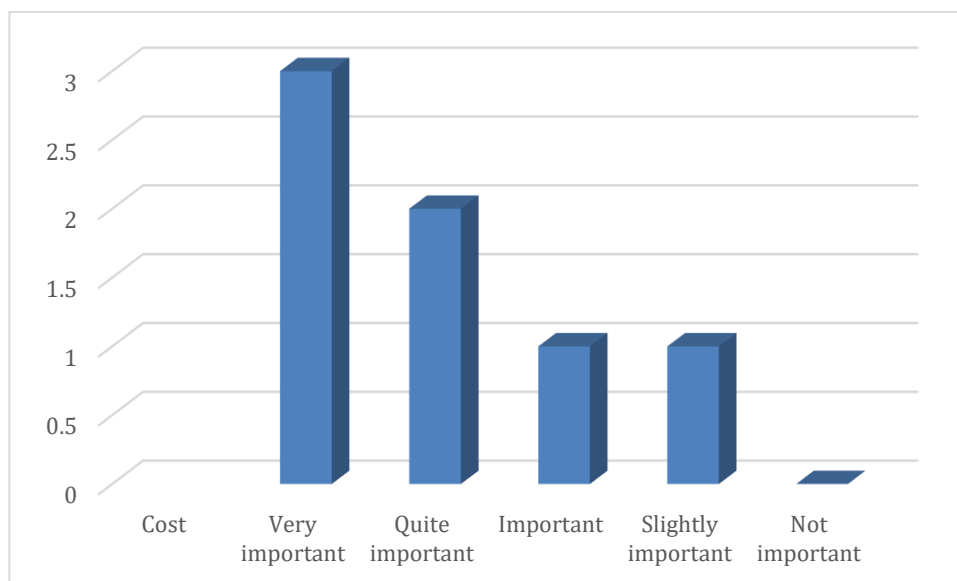


Figure 3.1-5 Cost as an important factor in energy source

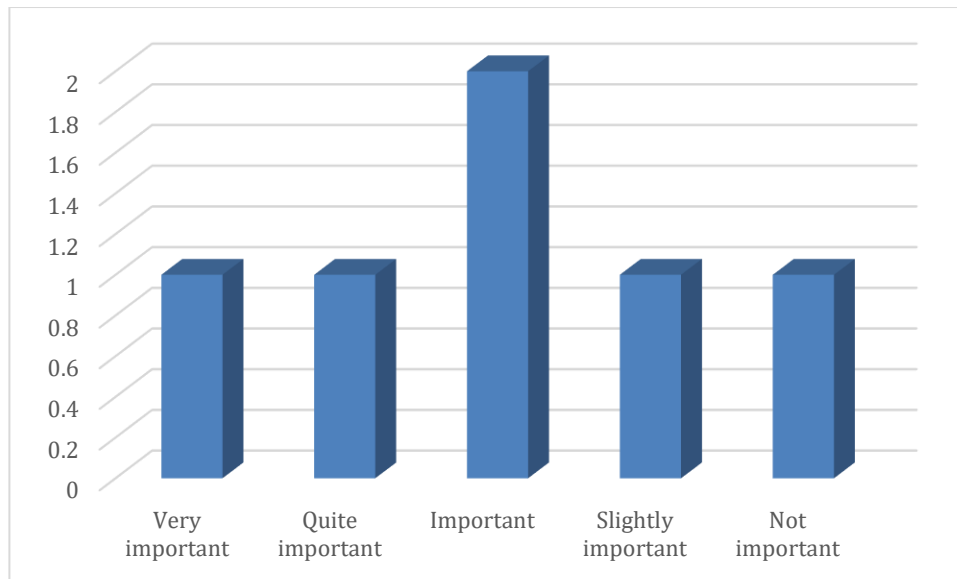


Figure 3.1-6Eco-friendly (low carbon footprint)

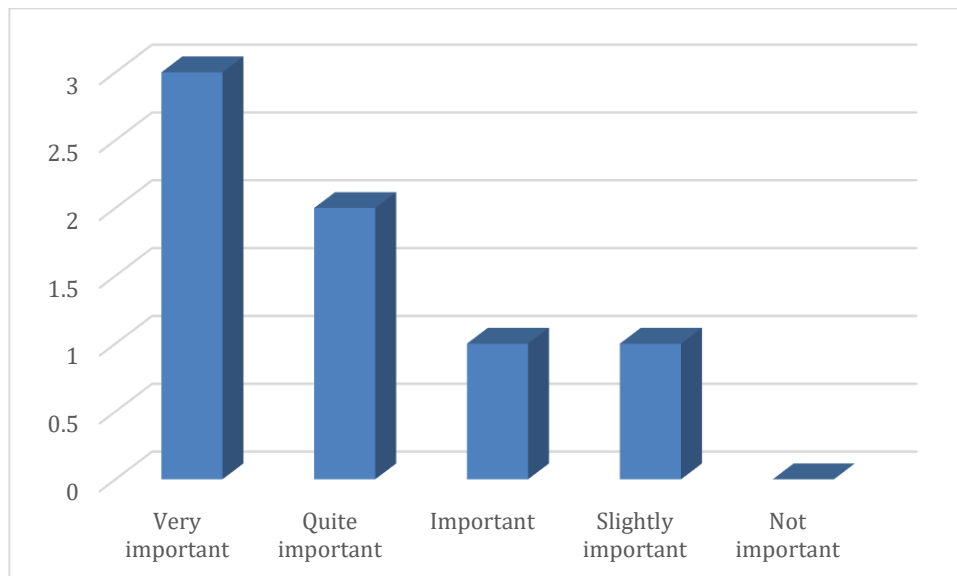


Figure 3.1-7Availability as an important factor in energy source

The respondents in the study all acknowledged the existence of environmental and green energy regulations (figure 7.1 -8).

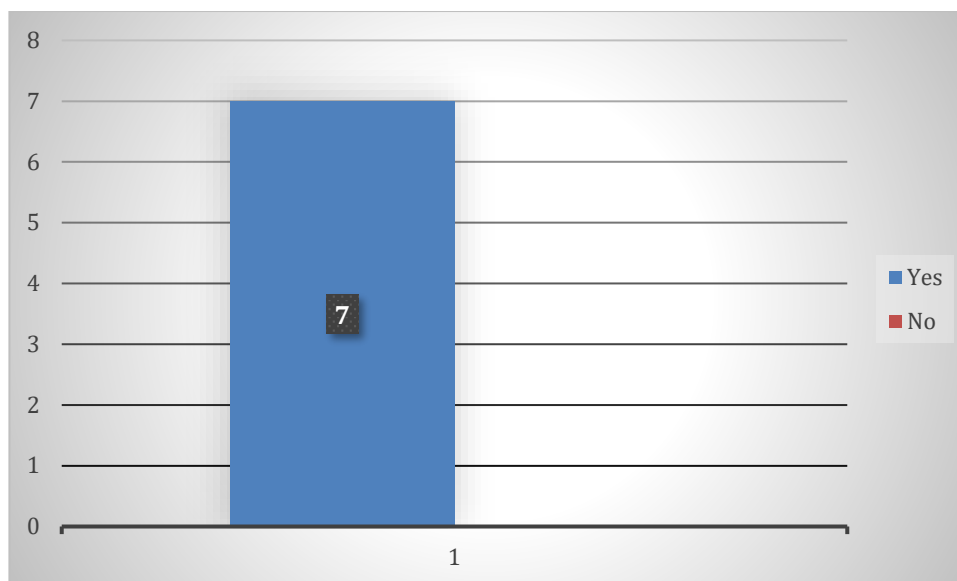


Figure 3.1-8 Environmental and Green Energy Regulations

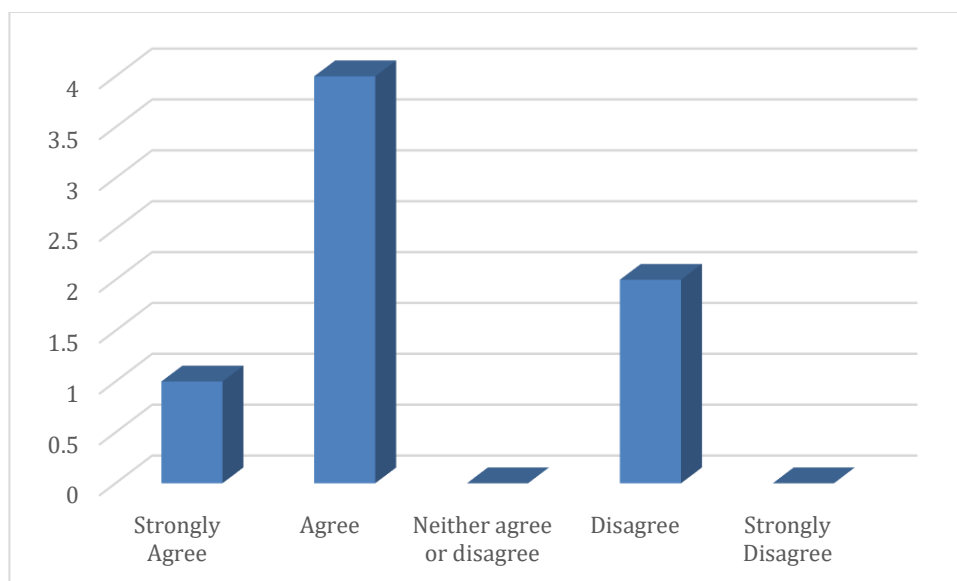


Figure 3.1-9 New policies have presented opportunities for the sector

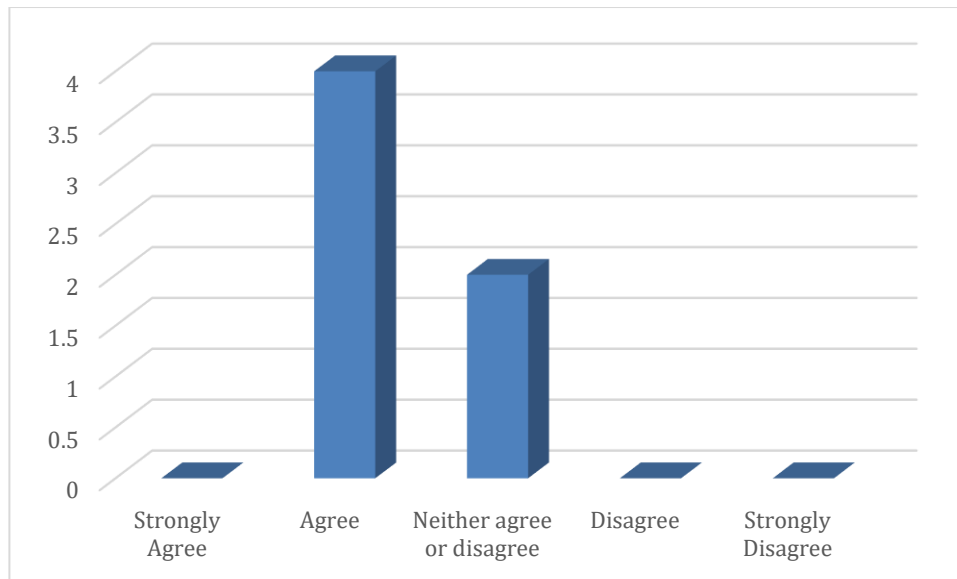


Figure 3.1-10 Information on current use efficiency practices is available

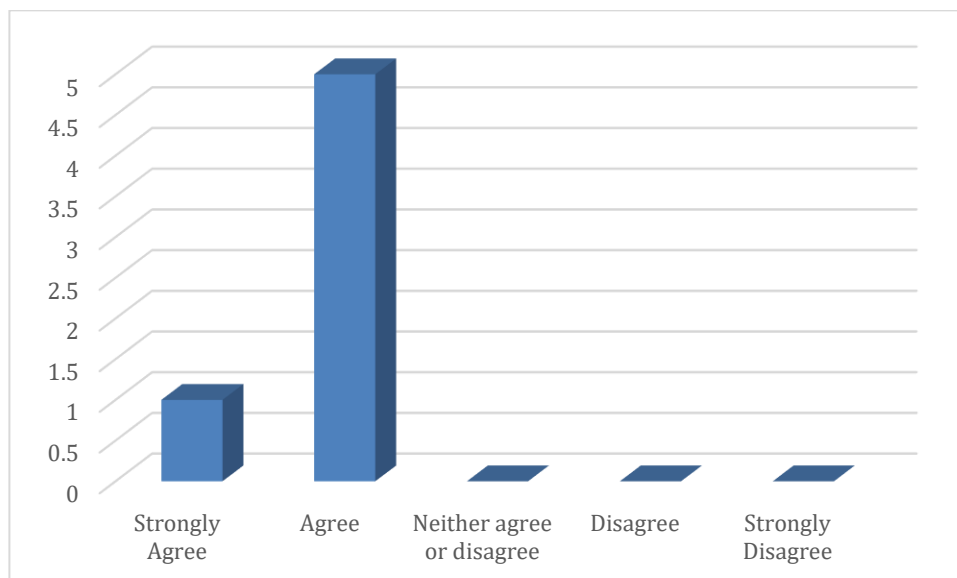


Figure 3.1-11 Technological trends in sector have a positive influence

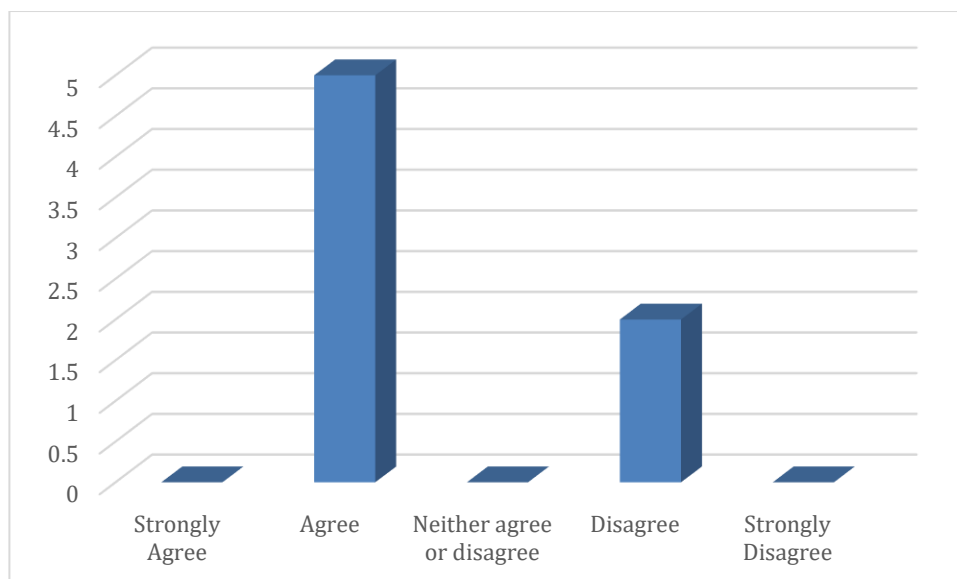


Figure 3.1-12 Incentives for green initiatives are available

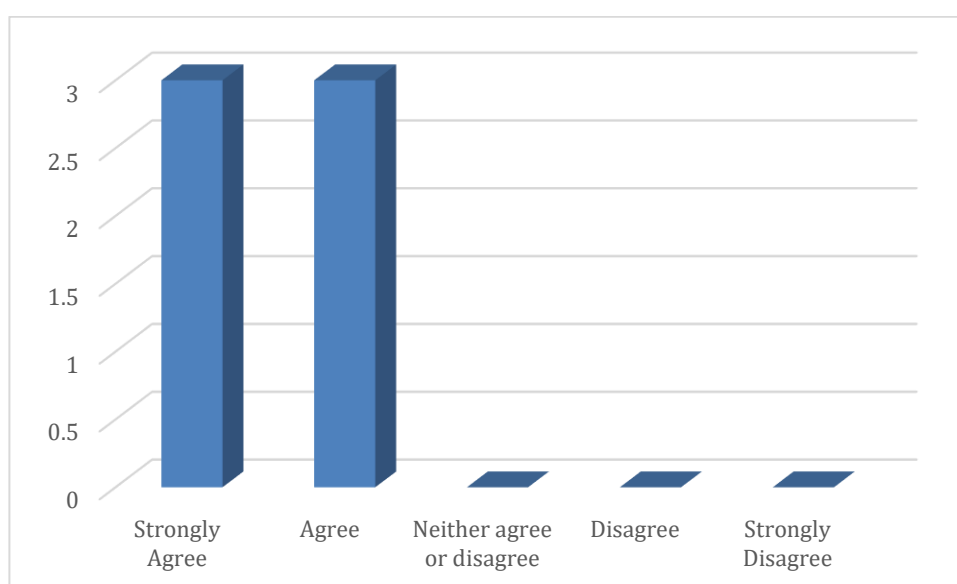


Figure 3.1-13 New technologies will result in new skills and competencies

THREATS

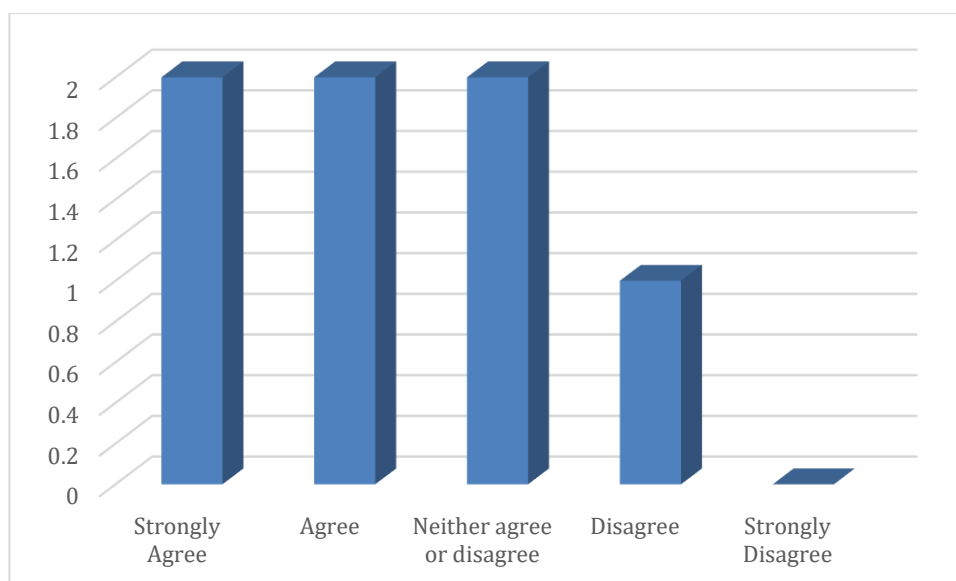


Figure 3.1-14 Natural factors-weather and climate change

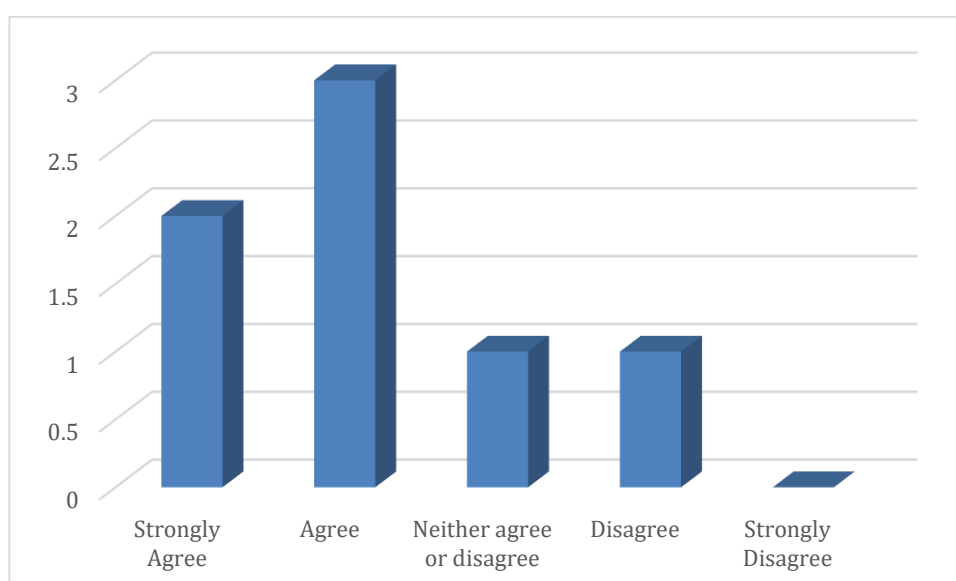


Figure 3.1-15 Sector products have a high environmental impact

3.2. Way forward in Green Economy planning

3.3. Environmental issues

As the Counties plan for mainstreaming GE, some major environmental issues were identified by the sectors studied. They have been classified into:

1. Land use
 - a. Extent and type of land cover,
 - b. Effective land management and sustainable use of natural resources (land and water),
 - c. Land degradation,

- d. Deforestation/catchment degradation,
 - e. Soil erosion and soil degradation,
 - f. Protection of riparian areas;
- 2. Pollution
 - a. Noise pollution,
 - b. Air pollution,
 - c. Pollution from use of agrochemicals,
 - d. Contamination of the ground hence effect on ecosystem;
- 3. Climate change
 - a. Erratic rainfall/droughts,
 - b. Decreased rainfall,
 - c. Reduced lake levels;
- 4. Waste Disposal
 - a. Small lagoons capacity,
 - b. Poor drainage systems,
 - c. Diminishing biodiversity.

3.4. Impacts of the environmental issues

The impacts of these issues were identified as:

- 1. Low land productivity,
- 2. increased vulnerability of communities,
- 3. Food insecurity,
- 4. Loss of property/ assets and loss of lives,
- 5. Soil erosion,
- 6. Increase of invasive species,
- 7. Climate change health hazards,
- 8. Livestock losses and loss of livelihood,
- 9. Contamination of water sources and air,
- 10. Conflicts and deaths.

3.5. Mitigating the impacts of the environmental issues

As we plan for Green Economy, mitigation is essential in an effort to reduce loss of life and property by lessening the negative impacts of disasters associated with current practices. In order for mitigation to be effective, we need to take action now, before the next disaster, to reduce human and financial consequences later. This entails issue risk analysis, reducing risk, and insuring against risk. It is important to know that disasters can happen at anytime and anyplace and if we are not prepared, consequences can be fatal.

Effective mitigation requires that stakeholders understand local risks, address the hard choices, and invest in long-term community well-being. Without mitigation actions, we jeopardize our safety, financial security and self-reliance. Some of the measures identified by the study include:

1. Awareness and enforcement of laws and regulations;
2. Training of farming communities to use appropriate technology in irrigation;
3. Training communities to conserve available water;
4. Providing effective legislation to check the phenomenon;
5. Carrying out routine environmental monitoring programme;
6. Setting up robust community relations departments;
7. Using improved silencers;
8. Moving forward with resettlement of communities;
9. Revegetating disturbed sites;
10. Conserving water catchments;
11. Rehabilitating forests;
12. Starting up income generating projects to reduce poverty;
13. Supporting communities with measures to tackle climate change issues;
14. Implementing community education, afforestation, and conservation initiatives.

3.6. Challenges to Green Economy implementation

The major Green Economy development challenges can be broadly similar across sectors and very particular at sector project level. Some of the challenges identified in this study include:

- High costs of investment in new processing equipment;
- Poor road network;
- Inadequate education and health care facilities and services;
- High unemployment among the youth, partly blamed for insecurity;
- Lack of legal framework for enforcement and funding limitations;
- Lack of awareness about good practice and technology by consumers;
- Competition for diminishing resources;
- Widespread poverty leads to overexploitation of resources;
- Low adaptation by stakeholders (attitudes are hard to change);
- Difficulties reaching out to the masses;
- Weak land tenure system;
- Rapid urbanization and housing problems;
- Perennial water shortages and growth of unplanned and informal settlements;
- High costs of rehabilitation of damaged ecosystems;
- Expensive alternative sources of fuel for some communities (Gas versus charcoal);
- Low funding in the sectors to enhance Green Economy initiatives;
- Difficulties to control what one does with their land;
- Corrupt personnel charged with taking care of forest resources.

3.7. Opportunities for Green Economy Implementation

There are several options which the County Government can take into consideration if they are serious about resolving the perennial garbage menace. They include:

- Repossess all the land which was grabbed or allocated fraudulently.
- Assess all the transit areas where garbage is collected, sorted and transported to different areas according to the kind of garbage.
- Offer the cleaning and waste management department adequate tools and train how to collect, sort and dispose garbage.
- Create awareness with the public and other stake holders to enhance and enforce on the importance and helpful measures of the 3R's: Recycle, Reuse and Reduce.
- Collect garbage which originates from the kitchens and dining tables indifferent containers and used for manure or animal feed.
- Provide portable containers/bins across the County in garbage collection with separate sections for different garbage collection.

The future of the waste management strategies have been summarized to include:

- Engage private facilities in management of solid waste at transfer stations;
- Increase rate of collection of solid waste at transfer points;
- Adopt new technologies in managing the waste;
- Invest in sanitary landfills.

4. County Planning Cycle

4.1. Institutional arrangements for annual development planning

There are a number of key documents, both at national and county level, that help to guide development planning at the county level. These include:

- The consultative Mombasa County Integrated Development Plan (CIDP) 2013-2017 where stakeholders gave their views in six stakeholder forums that were held in the constituencies of Likoni, Kisauni, Nyali, Mvita, Jomvu and Changamwe.
- The National Climate Change Response Strategy (NCCRS) 2010
- The County Government of Mombasa County Monitoring and Evaluation system, aligned to the National Integrated Monitoring and Evaluation system (NIMES), to serve the needs of the County. The system takes cognizance of the projects and programmes included in the County Integrated Development Plan as well as indicators facilitating the MTEF process and development of the country.

The following tools/methods/platforms are used by Planners:

- Tools
 - Monitoring committees, Master Plans, Strategic Plans
- Methods
 - Consultations, Active Engagement, SWOT, Cost Benefit Analysis
- Platforms
 - Communities, Public Participation

4.2. Relevant actors

	Sector	Actors
	Waste Management	Waste picker organizations within Mombasa include the Ministry of Youth and Sports and the Community Development Trust Fund. Their main source of funding is the community Development Trust Fund in Partnership with Destiny Africa and Ecotact.
	Cities	
	Transport	
	Fisheries	
	Energy	
	Water	
	Forestry	
	Buildings	
	Manufacturing	
	Agriculture	

4.3. Best entry point for the project in Mombasa County

A smooth transition to a Green Economy depends upon seven related conditions which are both necessary and sufficient to produce substantial changes, and which will enable governments to take on a more proactive role in reversing the course of carbon-based activities. These include:

political commitment, a legal and regulatory foundation, implementation of financial instruments, technological viability, human capital formation, appropriate institutional setup and a common language.

Political commitment to the green growth path presupposes that fiscal and monetary tools are used to steer economies away from business-as-usual. This would include investing in research and development (R&D), demonstration, deployment and the commercialization of different renewable technologies in production and consumption activities. For example, in the wake of the 1990s financial crises both Finland and Korea made strategic moves to increase R&D funding for renewables while scaling back on other public expenditures coupled with phasing out fuel subsidies². Currently, Finland and Korea have the competitive advantage in innovative technologies in renewable energies. Korea's New Green Deal established 17 new growth ventures and supported funds to research potential alternative energies. If decisions and policies are not anchored in a binding and enforceable framework, they become merely ornamental in nature. While we do not need to green all regulatory systems concurrently in order to respond to climate change challenges, legal standards must be assessed in a systematic approach. Legal reforms require revisiting existing laws and tuning their jurisdictions through amendments and directives. In some areas, this may require energizing the legal stand by conducting gap analysis, which entails comparing current environmental laws and procedures with international best practices in order to assess the current incentive system. The model of Kalundborg in Denmark demonstrates that wise planning can generate income and economic growth without adding environmental pressures. This distinctive example of using wastes from certain manufacturing operations as raw materials for industry was the result of coordinated efforts and a sustainability mindset in action.

Policy makers backed by a regulatory body have the power to initiate a paradigm shift using three simultaneous measures. Firstly, prioritise putting proper mechanisms in place to prevent natural resource use and abuse. This entails re-evaluation of the two types of resource usage, one being resource-use intensity and the other the depletion of natural stock. Secondly, Government bodies need to re-evaluate subsidy provisions and redirect funds towards ventures that provide permanent solutions to energy security issues. Finally, to reduce the financial risks associated with private green investment the Government should boost their R&D support, providing stimulus packages aimed at increasing the efficiencies of existing systems and supporting the development and demonstration stages of the renewable energy agenda.

Even with financial instruments in place, the choice of renewable technologies may prove to be a significant challenge. There are three steps by which choices in technology can be tested, the first of which is to use public funds to support working groups to debate the type of renewable technology most suited to current geopolitical and climatic conditions. The criterion of

² <http://www.greenecomonycoalition.org/know-how/seven-steps-towards-green-governance>

technology debates must be in line with technology suitability in order to meet the designated goal and to avoid duplications of efforts. Secondly, government funding for frequent sensitivity analysis will enable a clear vision of all options regarding renewable technologies. Thirdly, governments should champion resource-gap assessments. Such dialogues promote stakeholders to participate in finding solutions by acknowledging resource-gap difficulties. This gap analysis should be sponsored by different government bodies in coordination with practitioners, research groups and venture capitalists at the national and local levels.

Green Economy requires investments into human capital, a crucial expenditure that should not be compromised and, as part of public funding, ought to enhance the know-what, know-how and the know-why. Education, vocational training and research are extremely vulnerable areas during economic downturns. During recessions, governmental bodies (at both local and national levels) exercise greater budgetary discretion and often downsize publically funded activities such as education and training. Stripping educational budgets reduces the value of human capital that could otherwise help to stimulate the economy out of recession. It is during economic downturns that education and other human capital building vehicles need extra support to fund training and facilitate the formation of a cadre that fits the new green ideal. Lessons can be learned from Finland where during the economic crisis in the early 1990s, the government made a commitment to avoid cuts on essential services favouring R&D and educational institutions. Finland experienced a quick recovery from recession with a world-class educational system and highly skilled workforce.

Green Economy planning may require institutional adjustments for adoption and adaptation to innovative solutions, otherwise institutions will face “creative destruction” through inaction. The Counties may need to embrace protocols for local green initiative with modest goals to attract new investment and to bring in new revenue. The Counties can start by equipping new installations with solar panels, built biogas digesters and install windmills. Model gated communities may sell power back to the national grid as they generate more energy than they require.

References

- Abuodha, P.A.W., Kairo, J.G., 2001. Human-induced stresses on mangrove swamps along the Kenyan coast. *Hydrobiologia* 458, 255–265.
- Awuor, CB., Orindi, V.A. & Adwera, A.O., 2008. Climate change and coastal cities: the case of Mombasa, Kenya. *Environment & Urbanization* Vol 20(1): 231–242. DOI: 10.1177/0956247808089158
- Ballot, J., Hoyng, C., Kateman, I., Smits, M., de Winter, R., 2006. Report on coastal erosion project. Master's Thesis, Delft University of Technology, The Netherlands.
- Brown, S., Kebede, A.S., Nicholls, R.J., 2011. Sea Level Rise and Impacts in Africa, 2000 to 2100. School of Engineering and Environment, University of Southampton, Southampton, UK.
- Cinner, J.E., Daw, T.M., McClanahan, T.R., Muthiga, N., Abunge, C., Hamed, S., Mwaka, B., Rabearisoa, A., Wamukota, A., Fisher, E., Jiddawi, N., 2012. Transitions toward co-management: the process of marine resource management devolution in three east African countries. *Glob. Environ. Chang.* 22, 651–658.
- Commission on Revenue Allocation, 2013. Kenya County Fact Sheets, 2nd Edition.
- County Government of Mombasa, 2015a. Mombasa County Zoning Plan, 2015-2025. Department of Lands Planning and Housing.
- County Government of Mombasa, 2015b. Enhancing Economic Transformation for Shared Prosperity in Mombasa County. Fiscal Strategy Paper of the County Government of Mombasa, February 2015.
- FAO, 2007. Kenya Country Profile: Fisheries. FID/CP/KEN
- Githitho, A. 2004. The coastal terrestrial forests of Kenya. Report on resources threats and investments.
- Gitobu, J.K. & Njoroge, J.M., 2015. Adoption of green marketing practises by hotels in Mombasa County, Kenya. *African Journal of Tourism, Hospitality and Leisure Studies* Vol 1 (1) 1-18. AJTHLS - Open Access – Online @ <http://onlinesciencejournals.com/index.php/ajthls>
- Government of Kenya, 2009. State of the Coast Report. Towards Integrated Management of Coastal and Marine Resources in Kenya. National Environment Management Authority (NEMA), Nairobi. 88 pp.
- Government of Kenya, 2010. Shoreline Management Strategy for Kenya. National Environment Management Authority (NEMA), Nairobi, 87 pp.
- Hildebrand, M. and Holst, B. (2015). A Rapid Urban Diagnostic and Proposed Intervention Strategy for DIG in Mombasa, Kenya. http://www.urbisnetwork.com/documents/Mombasa_003.pdf
- IEA, 2015. Situational Analysis of Energy Industry, Policy and Strategy for Kenya. Published by the Institute for Economic Affairs, Nairobi, Kenya.
- Kairo, J.G., Bosire, J.O., Omar, M., 2005. Assessment of the Effects of Oil Spill on the Mangrove Forest of Port Reitz, Mombasa, 12 pp.

- Kairu, K., Nyandwi, N. (Eds.), 2000. Guidelines for the study of shoreline change in the Western Indian Ocean region. IOC Manuals and Guides No. 40. UNESCO 2000.
- Kebede, A.S., Hanson, S., Nicholls, R.J., Mokrech, M., 2009. Impacts of Climate Change and Sea-Level Rise: A Case Study of Mombasa, Kenya. School of Engineering and Environment, University of Southampton, Southampton, UK.
- Kenya Water Report, 2005. Water for Wealth Creation and Healthy Environment for a Working Nation. UN-WATER/WWAP/2006/12.
- KNBS, 2014. Kenya Facts and Figures.
- McClanahan, T.R., 1997. Primary succession of coral-reef algae: differing patterns on fished versus unfished reefs. *J. Exp. Mar. Biol. Ecol.* 218, 77–102.
- Ministry of Fisheries Development, 2012. Marine Waters Fisheries Frame Survey 2012 Report.
- Ministry of Health, 2014. County Sanitation Profiles - Water and Sanitation Programme (WSP). <https://www.wsp.org/sites/wsp.org/files/publications/WSP-Kenya-47-County-Sanitation-Profiles-2014.pdf>
- Mombasa CIDP (2013). First County Integrated Development Plan 2013-2017. Mombasa County Government.
- Mombasa County Annual Development Plan, 2014. Theme: A Globally Competitive and Prosperous Kenya. Mombasa County Annual Development Plan 2015-2016 Fiscal Year. Budget Office of the County Government of Mombasa, Finance and Economic Department.
- Msengeti, DM. & Obwogi, J., 2015. Effects of Pay and Work Environment on Employee Retention: A Study of Hotel Industry in Mombasa County. *International Journal of Scientific and Research Publications*, Volume 5, Issue 4, April 2015, 10pp.
- Muohi, A.W., Onyari, J.M., Omondi, J.G., Mavutia, K.M., 2003. Heavy metals in sediments from Makupa and Port–Reitz Creek systems Kenyan Coast. *Environ. Int.* 28, 639–647.
- Murai, CB & Kirima, W. 2015. Aligning Kenya's Financial System with Inclusive Green Investment. Current Practice and Future Potential to Mobilize Investment in a Sustainable Economy. International Finance Corporation and United Nations Environment Programme, 2015.
- Muthiga, N.A., 2009. Evaluating the effectiveness of management of the Malindi–Watamu marine protected area complex in Kenya. *Ocean Coast. Manag.* 52, 417–423.
- Ngugi, E., 2013. Exploring Kenya's Inequality: Pulling Apart or Pooling Together? Mombasa County. Published by Kenya National Bureau of Statistics (KNBS) and Society for International Development (SID), 73pp.
- Njoroge, J.M. 2015. Climate change- perceived impacts, risks, vulnerability, and response strategies: A case study of Mombasa coastal tourism, Kenya. *African Journal of Hospitality, Tourism and Leisure* Vol. 4 (1) - (2015) AJHTL - Open Access- Online @ [http://: www.ajhtl.com](http://www.ajhtl.com)
- Obura D, Celliers L, Machano H, Mangubhai S, Mohammed M, Motta H, Muhando C, Muthiga N, Pereira M, Schleyer M (2002). Status of Coral reefs in Eastern Africa: Kenya, Tanzania, Mozambique and South Africa. In: Wilkinson C (ed) *Status of Coral Reefs of the World*, 2002. Global Coral Reef Monitoring Network (GCRMN). Australian Institute of Marine Science, Townsville, Australia, p 63-78

- Odada, E.O., 1995. Coastal Erosion and Flooding in East Africa and the Western Indian Ocean. In: Proceedings of the Workshop and Policy Conference on Integrated Coastal Zone Management in Eastern Africa Including the Island States, 21–13 April 1993, ISBN: 971-9014-00-8.
- Odero, W.O. & Reeves, W. 2014. Kenya 2014. Africa Environment Outlook. AfDB, OECD, UNDP 2014. www.africaneconomicoutlook.org
- Ogada, M.O. and Nyingi, D.W., 2013. The Management of Wildlife and Fisheries Resources in Kenya: Origins, Present Challenges and Future Perspectives. IN: Paolo Paron, Daniel Olago and Christian Thine Omuto, editors: Developments in Earth Surface Processes, Vol. 16: 219-236. Amsterdam, The Netherlands.
- Omuombo, CA, Olago, DO, Odada, EO (2013a) Coastal Erosion. IN: Paolo Paron, Daniel Olago and Christian Thine Omuto, editors: Developments in Earth Surface Processes, Vol. 16: 331-339. Amsterdam, The Netherlands.
- Omuombo, CA, Olago, DO, Odada, EO (2013b) Coastal Waters. IN: Paolo Paron, Daniel Olago and Christian Thine Omuto, editors: Developments in Earth Surface Processes, Vol. 16: 109-114. Amsterdam, The Netherlands.
- Ongoma V, Onyango OA (2014) A Review of the Future of Tourism in Coastal Kenya: The Challenges and Opportunities Posed by Climate Change. J Earth Sci Clim Change 5: 210. doi:10.4172/2157-7617.1000210
- Owinga, T. & Datche, E., 2015. Strategic Factors Leading to the Growth of Tourism in Mombasa County. International Journal of Science and Research (IJSR) Volume 4 Issue 8, August 2015. 10pp.
- Owuor, G.G, 2011. Assessment of sub-surface hydrocarbon contamination resulting from fuel storage depots, Shimanzi area, Mombasa, Kenya. MSc. Thesis, Department of Geology, University of Nairobi, Nairobi, Kenya. 72pp.
- Palfreman, J. & Clark, M., 2015. Mapping out waste characteristics in Mombasa, Kenya. 15th International Waste Management and Landfill Symposium, S. Margherita di Pula, Cagliari, Italy, 5-9 October 2015. CISA Publisher, Italy. <http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya>
- Rakodi, C., Gatabaki-Kamau, R. and Devas, N., 2000. Poverty and political conflict in Mombasa. Environment&Urbanization Vol 12 No 1, pp. 153-170.
- Steyn, G. (2015) 'The Impacts of Islandness on the Urbanism and Architecture of Mombasa', Urban Island Studies, 1, 55-80.
- UNEP. (2015). Green Economy Scoping Study: Kenya.
- Williams, T.M., Rees, J.G., Ferguson, A., Herd, R.A., Kairu, K.K., Yobe, A.C., 1997. Metals, petroleum hydro-carbons and organochlorines in inshore sediments and waters of Mombasa, Kenya. Mar. Pollut. Bull. 34 (7), 570–577.
- World Bank 2012. *What a Waste - A Global Review of Solid Waste Management 2012*. Coast Water Services Board's (CWSB) report for 2012.
- WRI, 2007. Nature benefits in Kenya: an atlas of ecosystem and human well-being. World Resource Institute, Department of Resource Surveys and Remote Sensing, Ministry of Environment and

Natural Resource, Kenya Central Bureau of Statistics, Ministry of Planning and National Development, Kenya and International Livestock Research Institute.